

Private Proofs of When and Where

Feasibility and Applications of Zero-Knowledge Position Verification

Leo Orshansky (Columbia University)

To appear at CRYPTO 2026!



Uma Girish

Columbia



Grzegorz Gluch

UC Berkeley/MIT



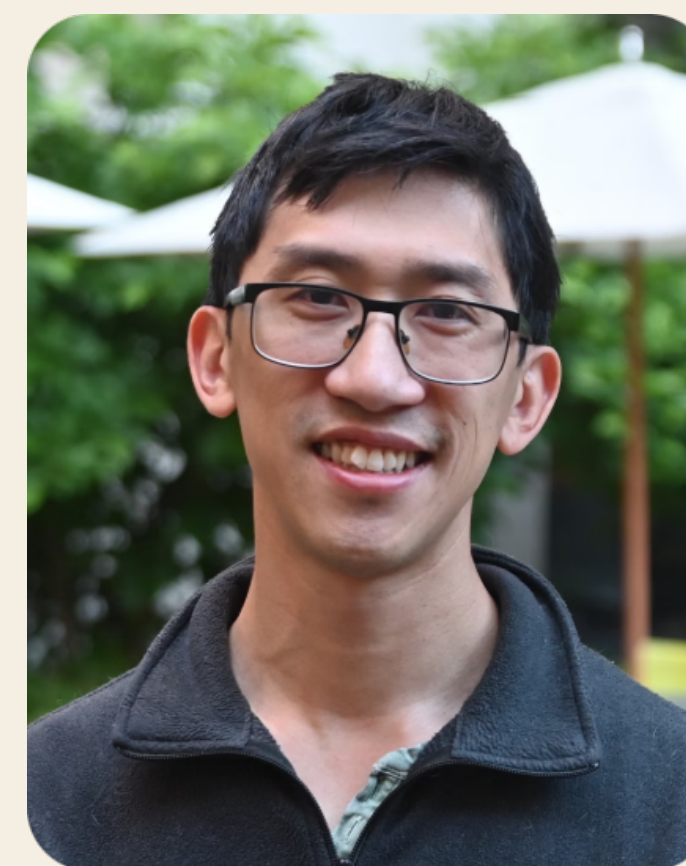
Shafi Goldwasser

UC Berkeley/MIT



Tal Malkin

Columbia

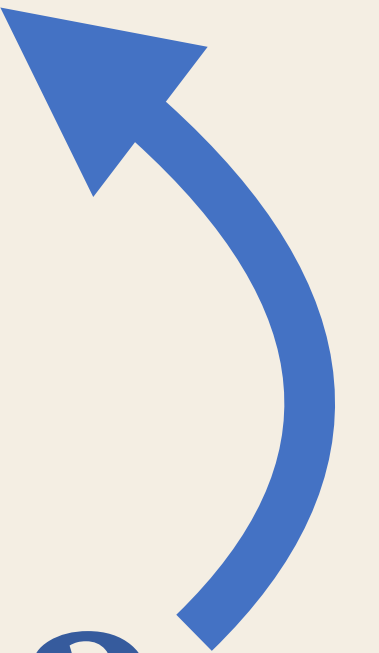


Henry Yuen

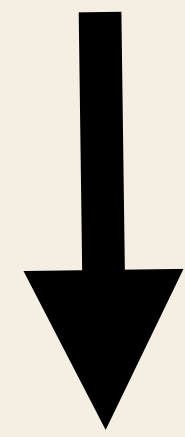
Columbia

Private Proofs of When and Where

Part One: What on earth are those?

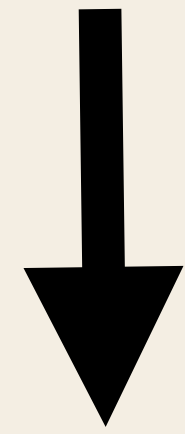


Private Proofs of When and Where



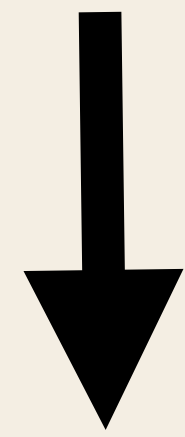
Peter to Veronica: "I can convince you that..."

Private Proofs of When and Where

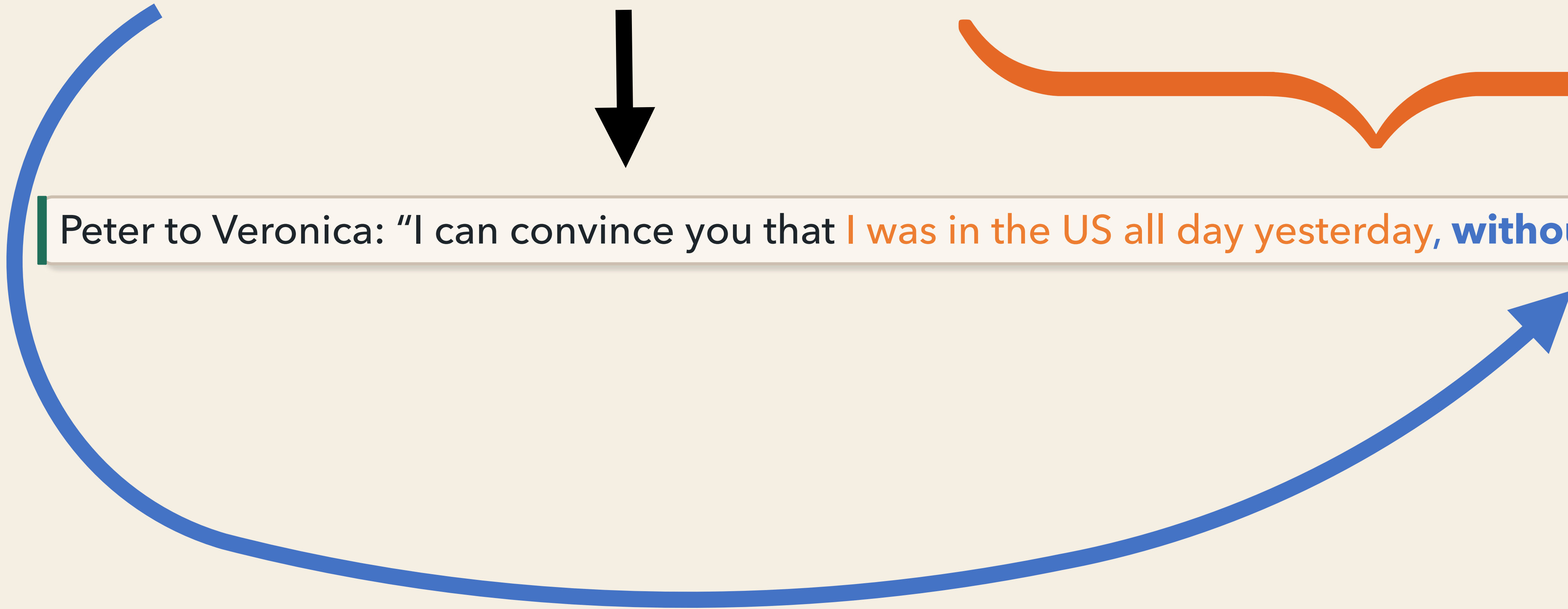


Peter to Veronica: "I can convince you that I was in the US all day yesterday"

Private Proofs of **When and Where**



Peter to Veronica: "I can convince you that I was in the US all day yesterday, **without saying where.**"



Scenario: Peter's Secret Lookout

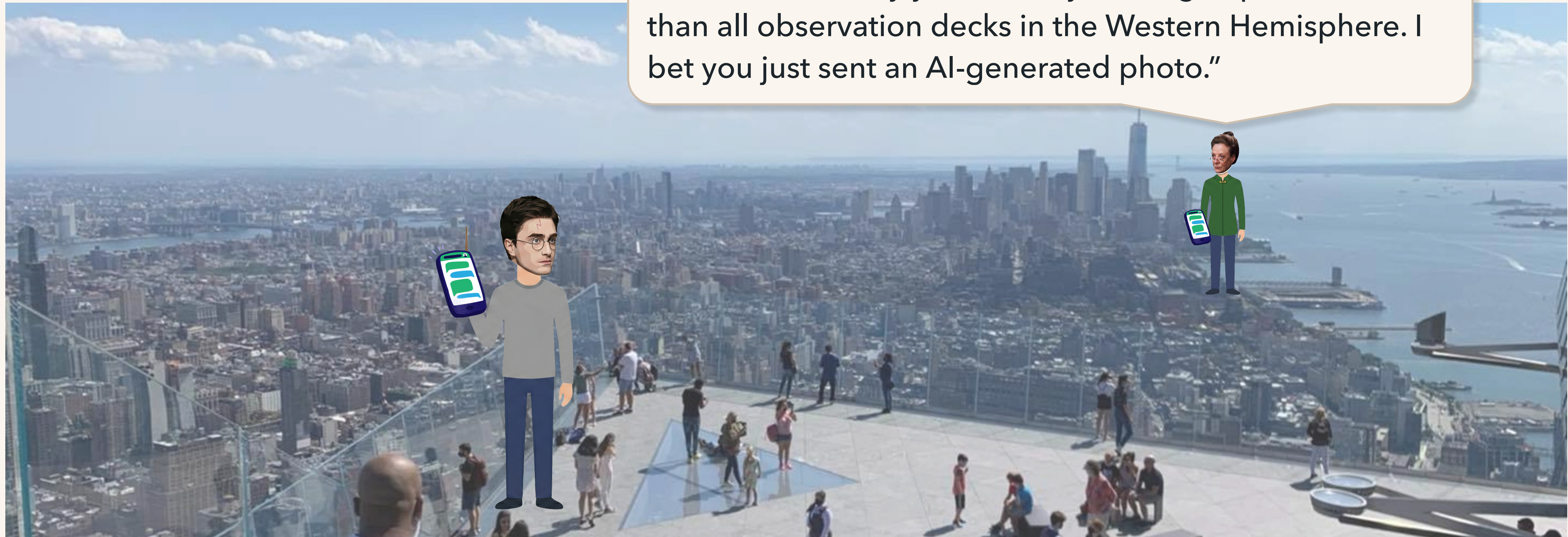
Peter finds a lookout spot 350m above NYC. How can he prove it to Veronica?



Scenario: Peter's Secret Lookout

Peter finds a lookout spot 350m above NYC. How can he prove it to Veronica?

Veronica: "No way you're really that high up. That's taller than all observation decks in the Western Hemisphere. I bet you just sent an AI-generated photo."



Scenario: Peter's Secret Lookout

Peter finds a lookout spot 350m above NYC. How can he prove it to Veronica?



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Peter finds a lookout spot 350m above NYC. How can he prove it to Veronica?



Position as a Credential

Shifting identity from *who* you are to *where* you are

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“Who you are” – AKA what secret information do you know?

- Bank account password
- PKE decryption key
- Digital signature signing key

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- Access to a secure facility
- Located within a particular jurisdiction

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A new kind of cryptography: **position-based cryptography**

Position-Based Cryptography

Position-Based Authentication

Peter wants to authenticate some message m as originating from location L .

1. If Peter is indeed at location L ,
Veronica accepts and outputs m .
2. If Peter is not at location L ,
Veronica rejects w.h.p.

Position-Based Cryptography

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Position-Based Key Exchange

Veronica wants to communicate secretly with whomever is at location L .

1. If Peter is at location L , he and Veronica establish a shared key.
2. For anyone not at location L , the key is completely hidden.

How to build position-based crypto?

This Work:

Private Proofs of When and Where

This Work:

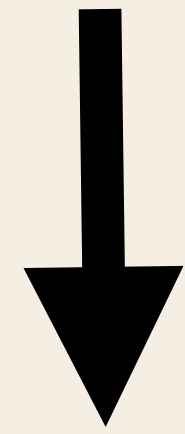
Private Proofs of ~~When and Where~~

(Prior Work)

Proofs of Here and Now

(Prior Work)

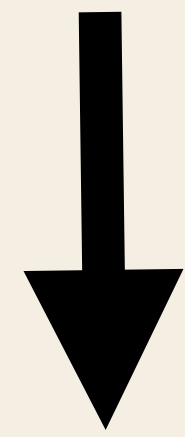
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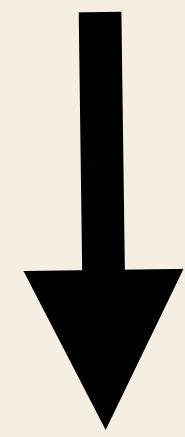
Proofs of Here and Now



Peter to Veronica: "I can convince you that I am currently located at coordinates XYZ."

(Prior Work)

Proofs of Here and Now

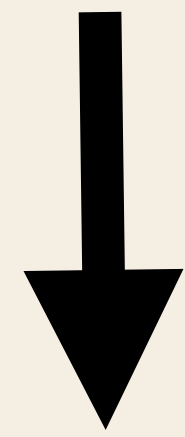


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This problem is called **Position Verification**

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Proofs of Here and Now



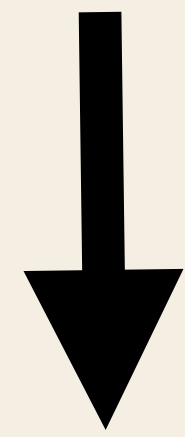
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[CGMO'09]: Position verification -> Position-based key exchange

(Prior Work)

Proofs of Here and Now



Peter to Veronica: "I can convince you that I am currently located at coordinates XYZ."

This problem is called **Position Verification**

[CGMO'09]: Position verification -> Position-based key exchange

[BCFGGOS'10, U'14]: Position verification -> Position-based authentication (bonus: -> QKD)

But wait: isn't this just **GPS**?

GPS cannot help Veronica tell whether Peter is *lying* about his location.

GPS

- Great for finding out one's *own* location
- Fully passive: satellites broadcast, but receiver just listens

Proof of position

- Verifier attempts to learn the true location of the prover
- If the prover is lying, the interaction should fail w.h.p

Need *interaction* to catch dishonest provers.

Position Verification

The central building block of position-based cryptography.



Location L



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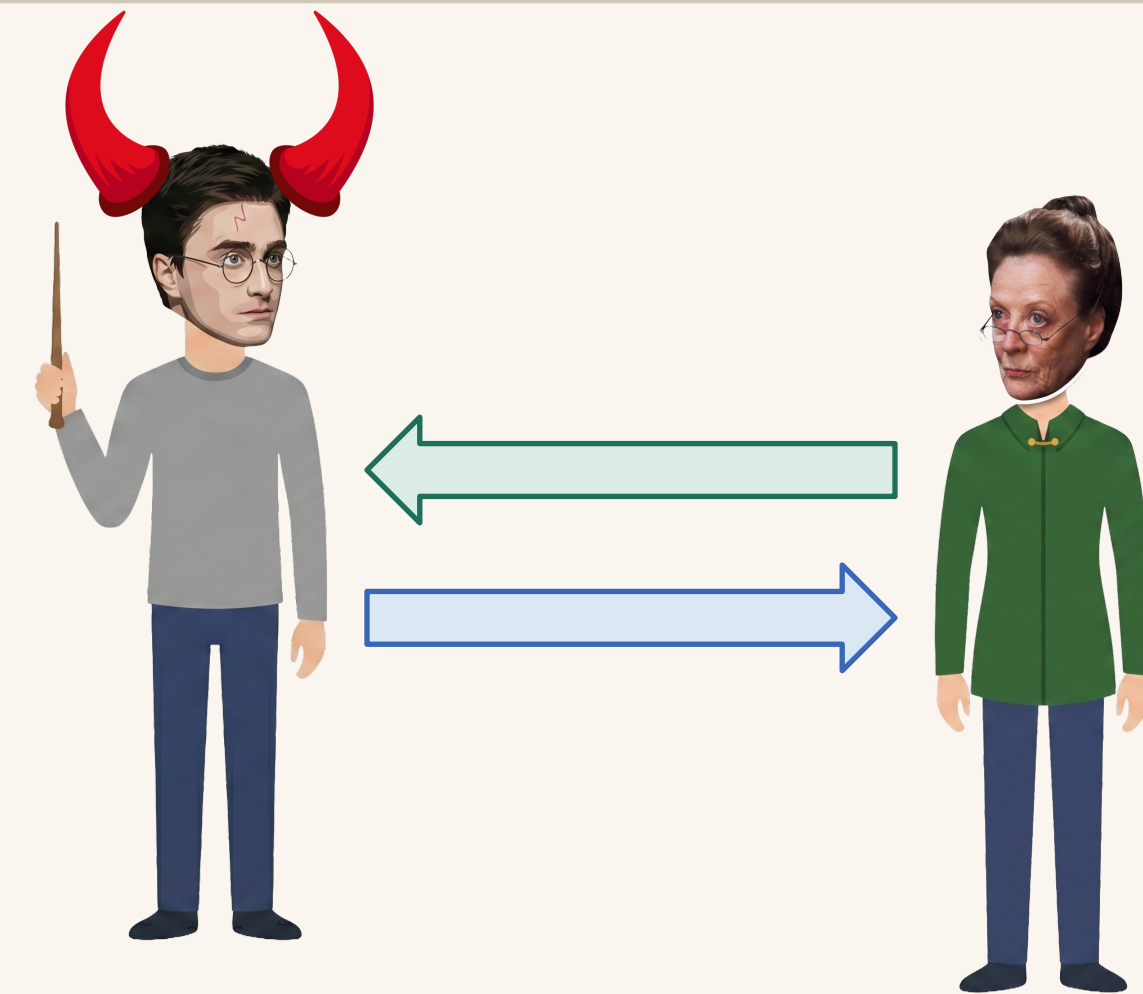


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The central building block of position-based cryptography.

What if Peter tries to spoof?

Location L



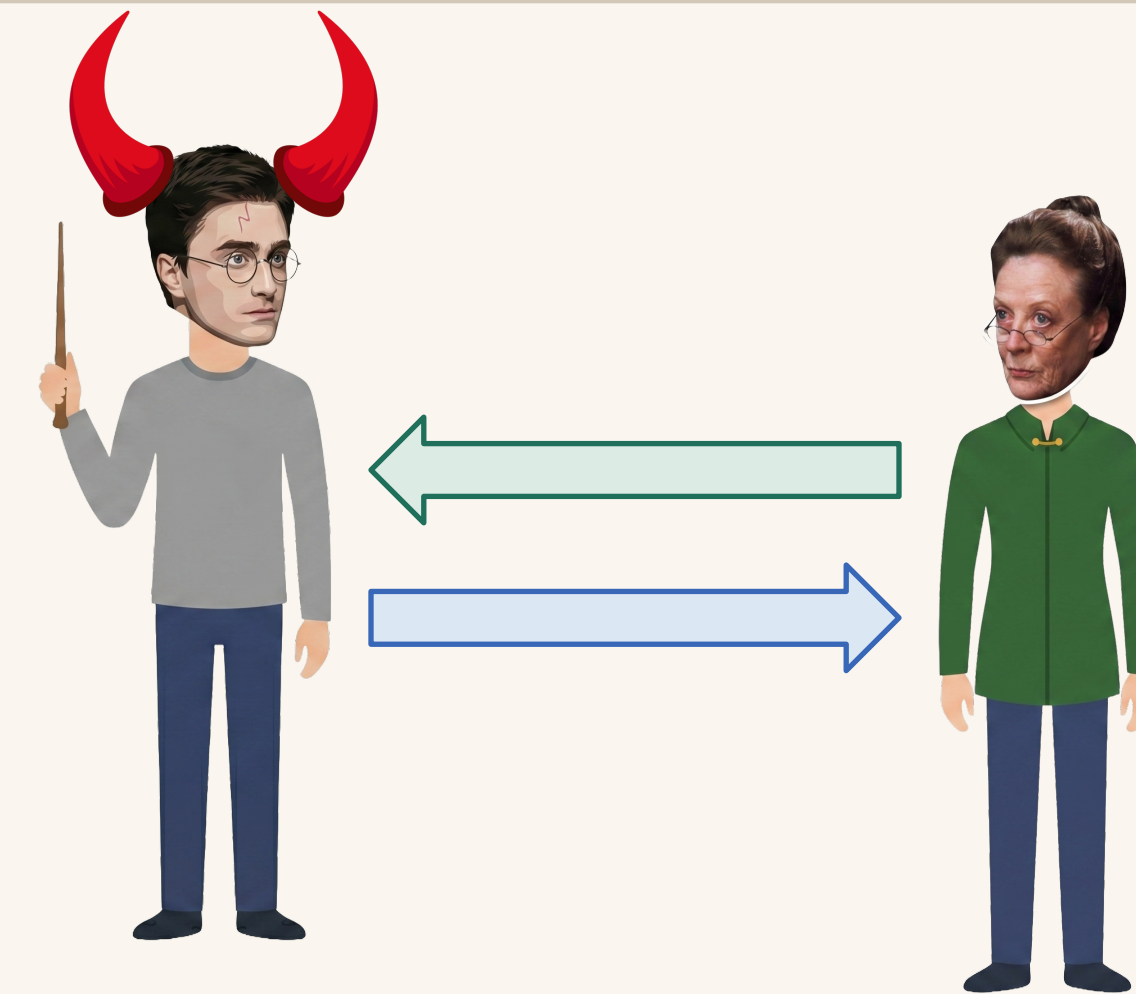
Position Verification

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He has the same view as he would if he
was at location L...

Can produce the exact same responses.

Location L



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Position Verification

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Victoria

Add second verifier:
Victoria

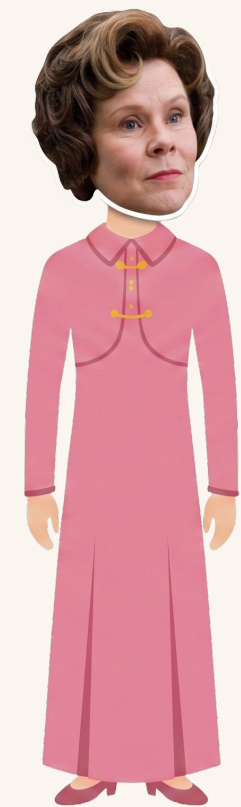


Location L

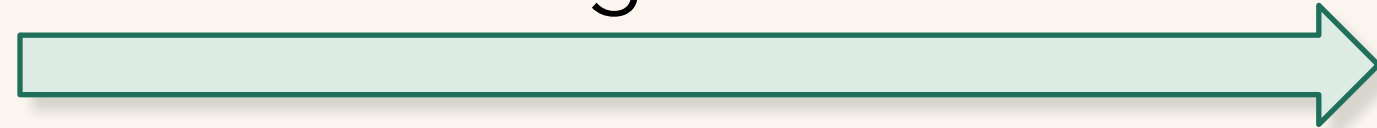


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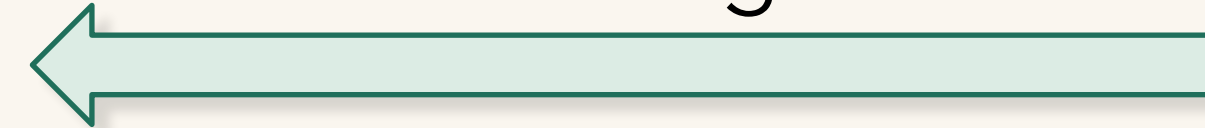


Challenge 1



Location L

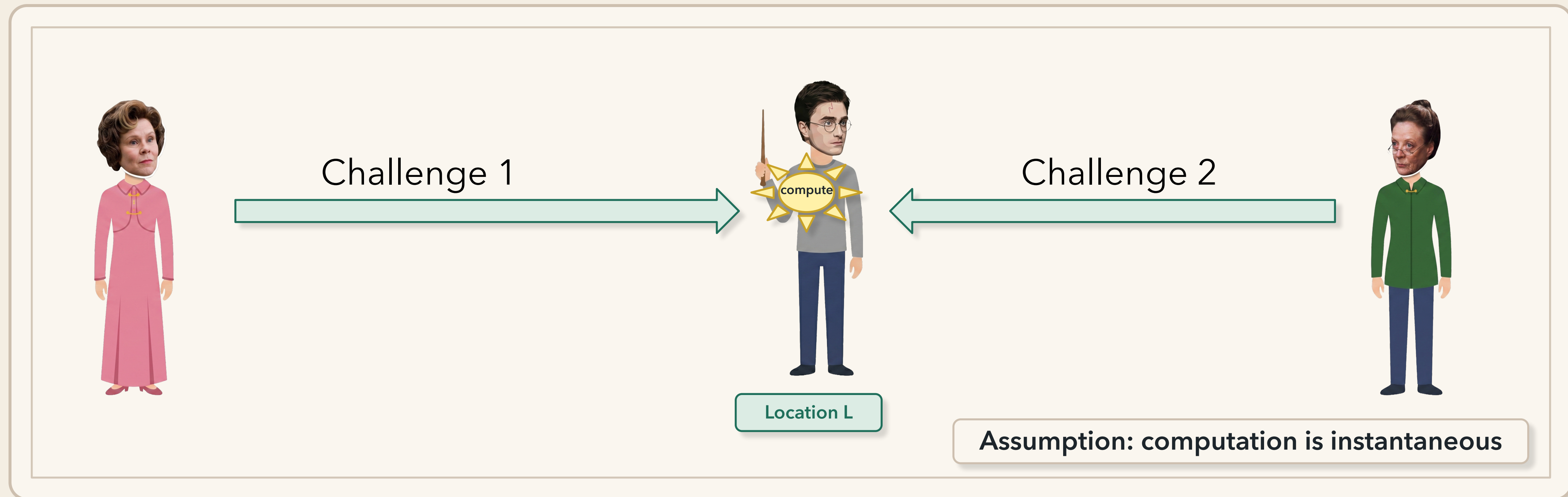
Challenge 2



Assumption: messages travel at light speed

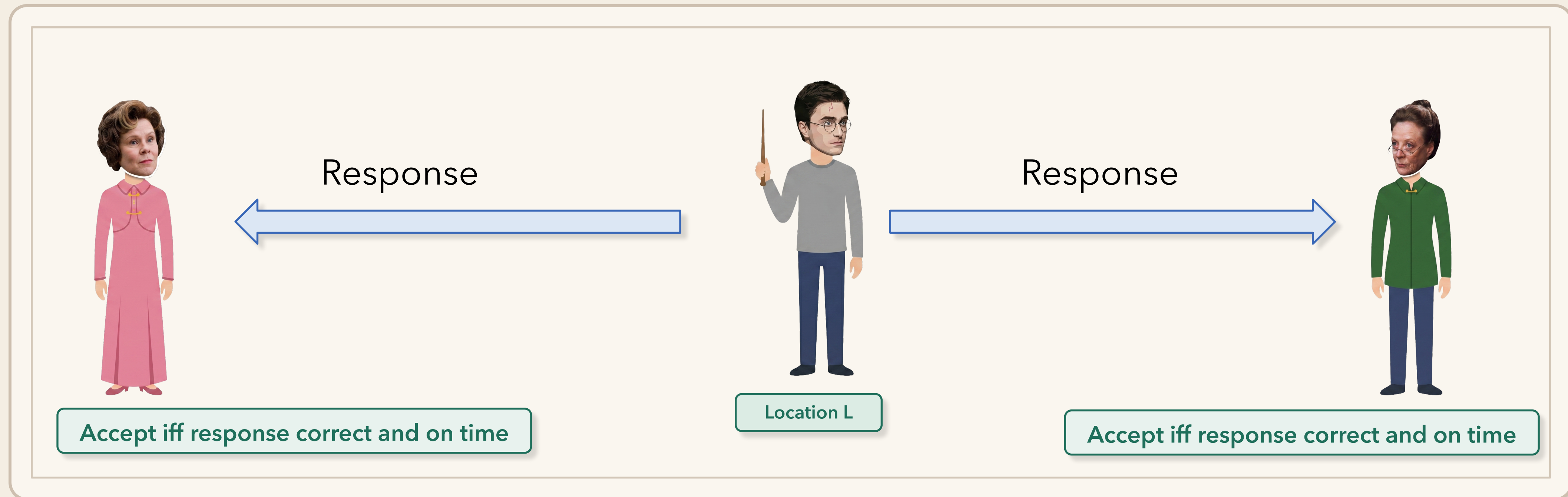
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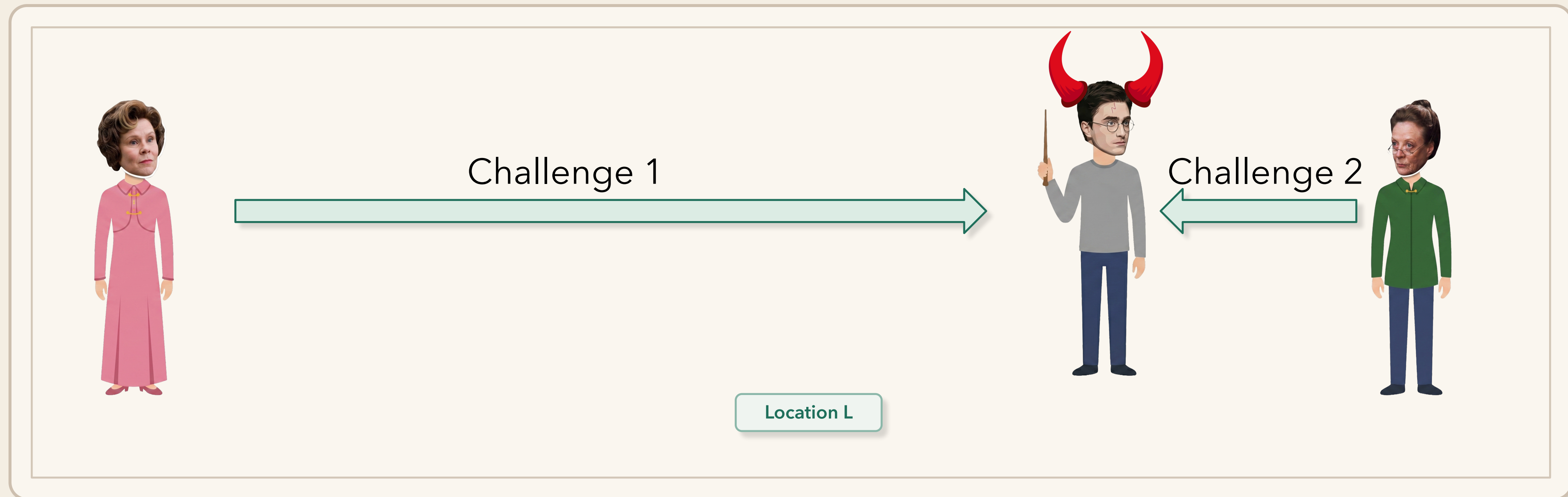
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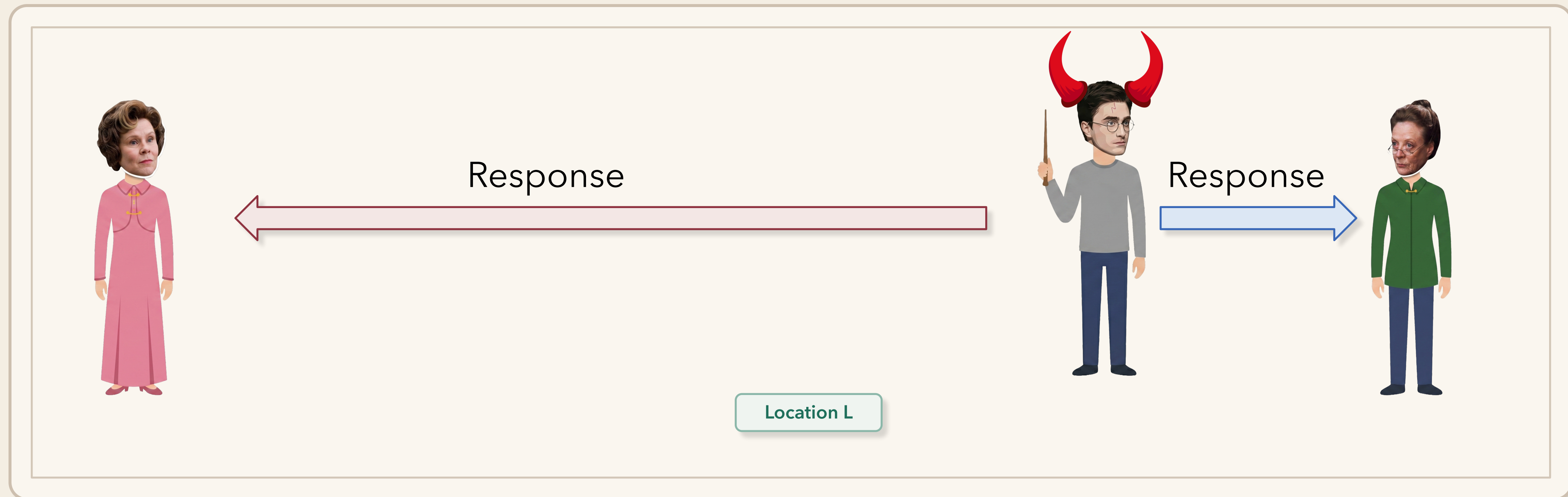
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Secure against spoofing prover!

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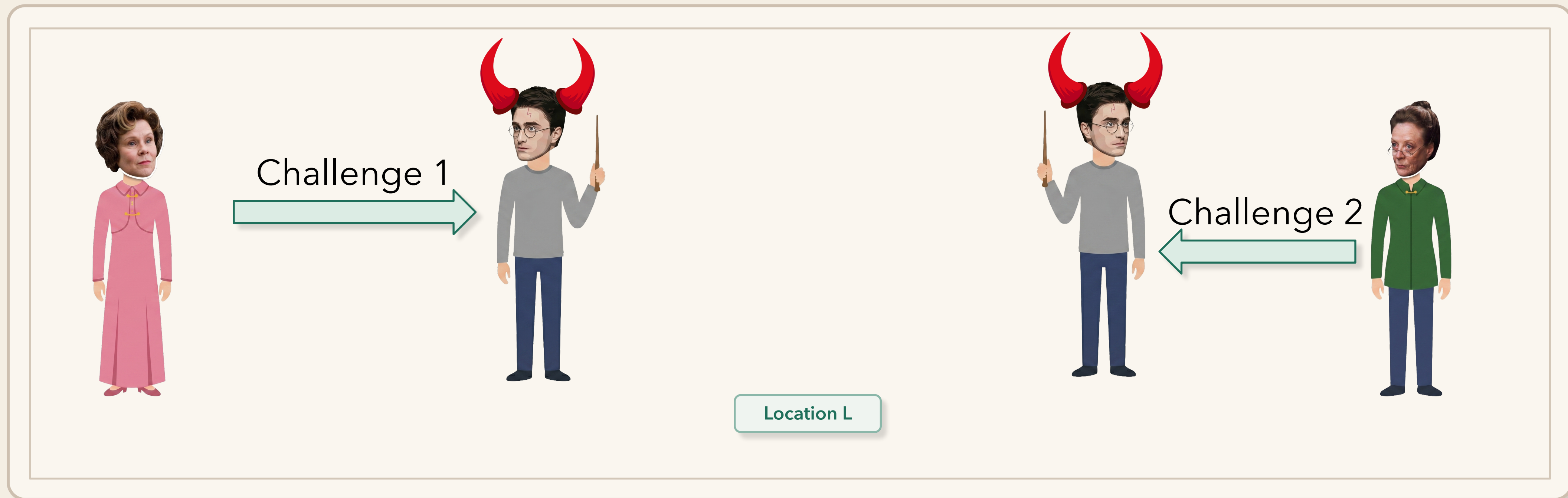
What if Peter gets his friend to help him spoof?



Location L

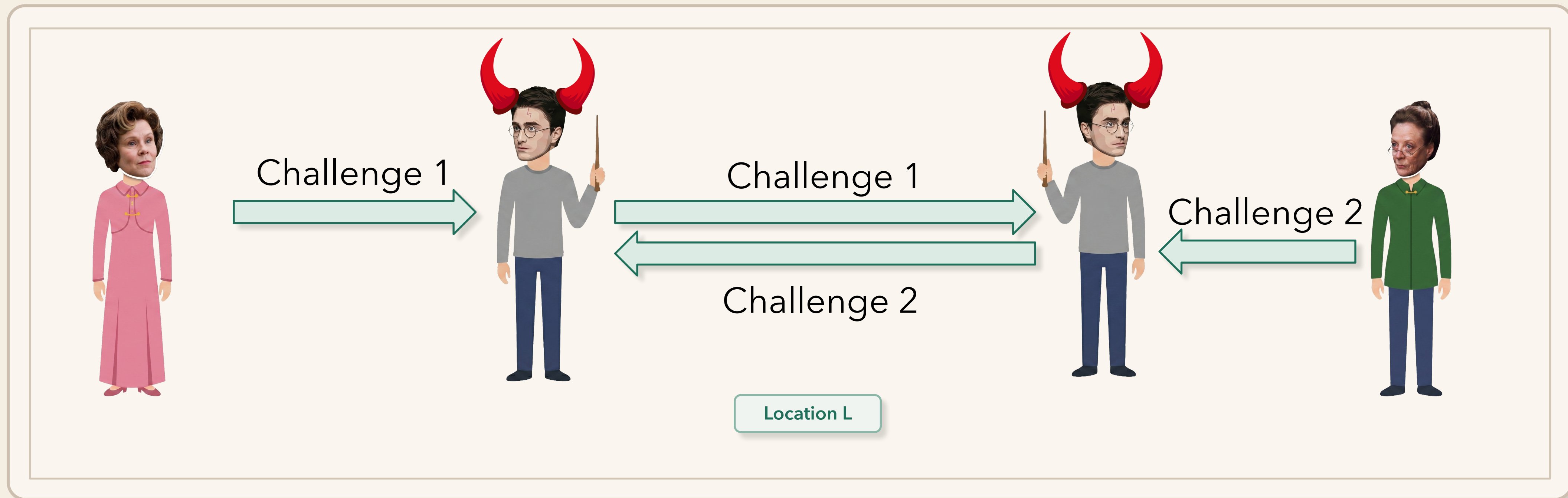
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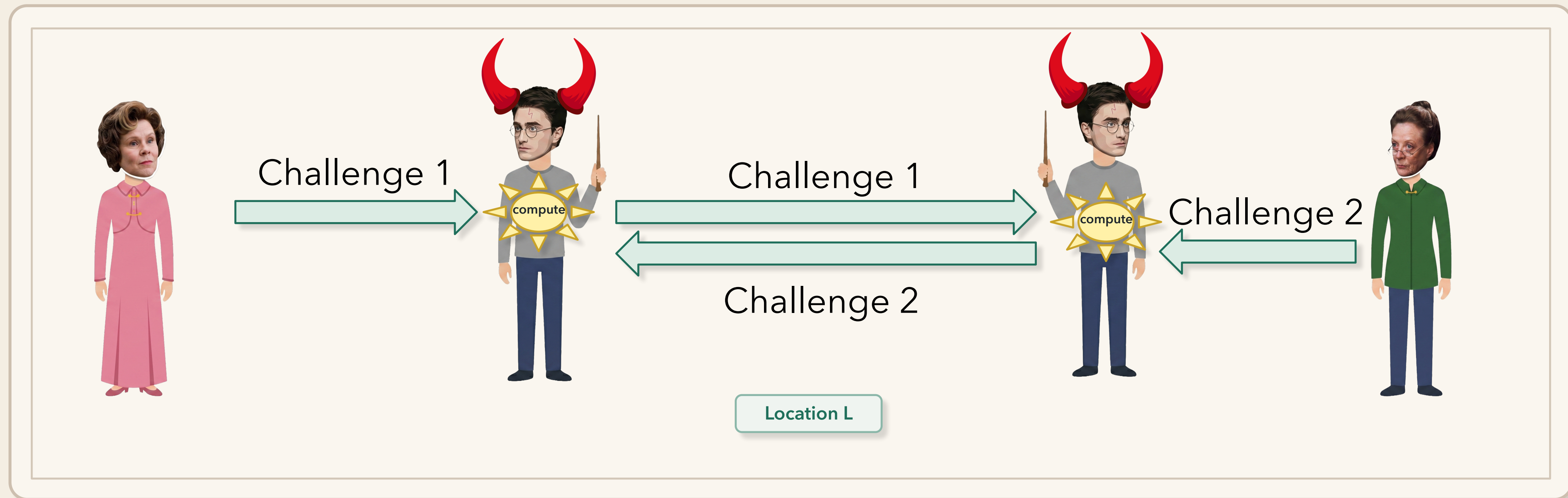
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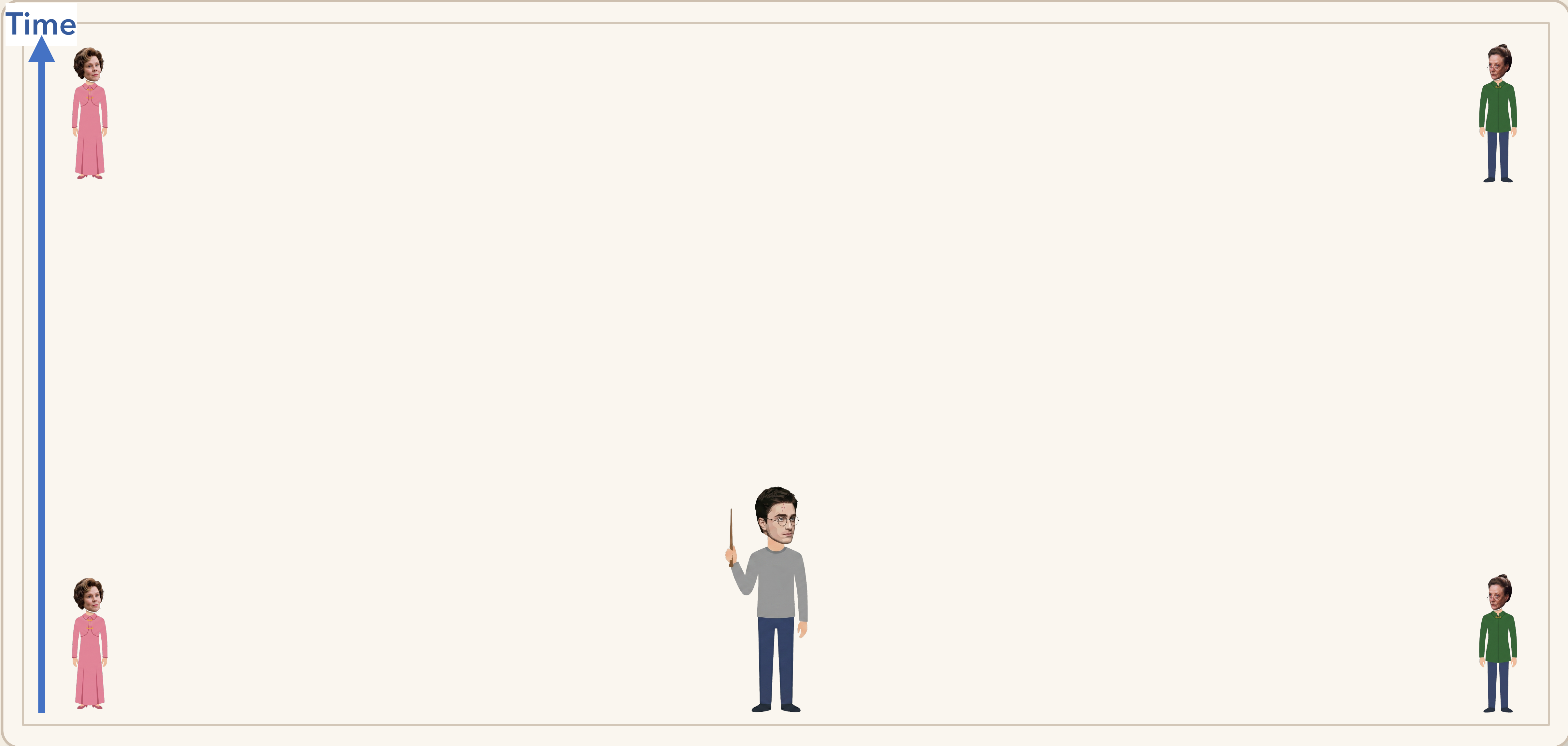
Spoofing provers each get the honest prover's view, just in time....

Position Verification

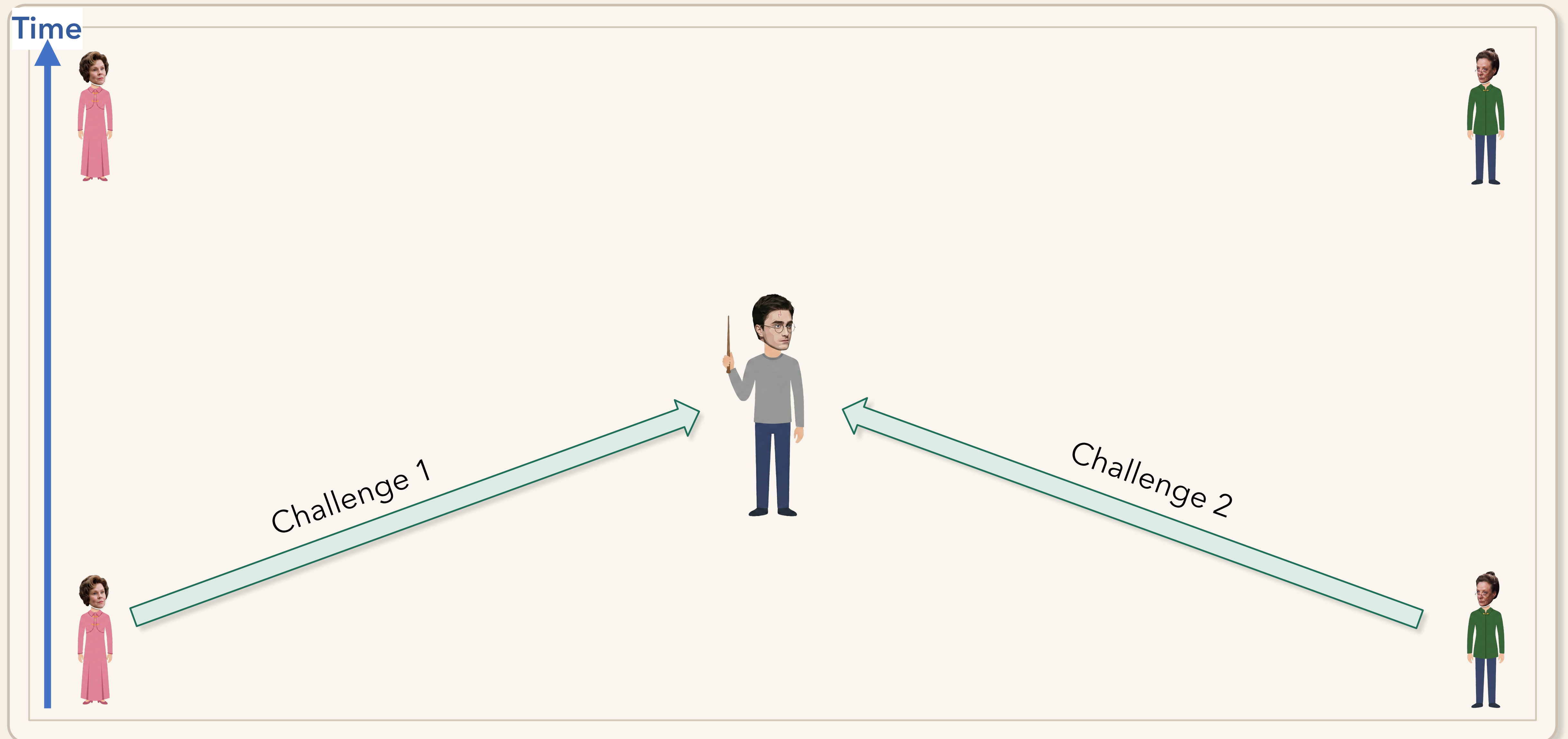
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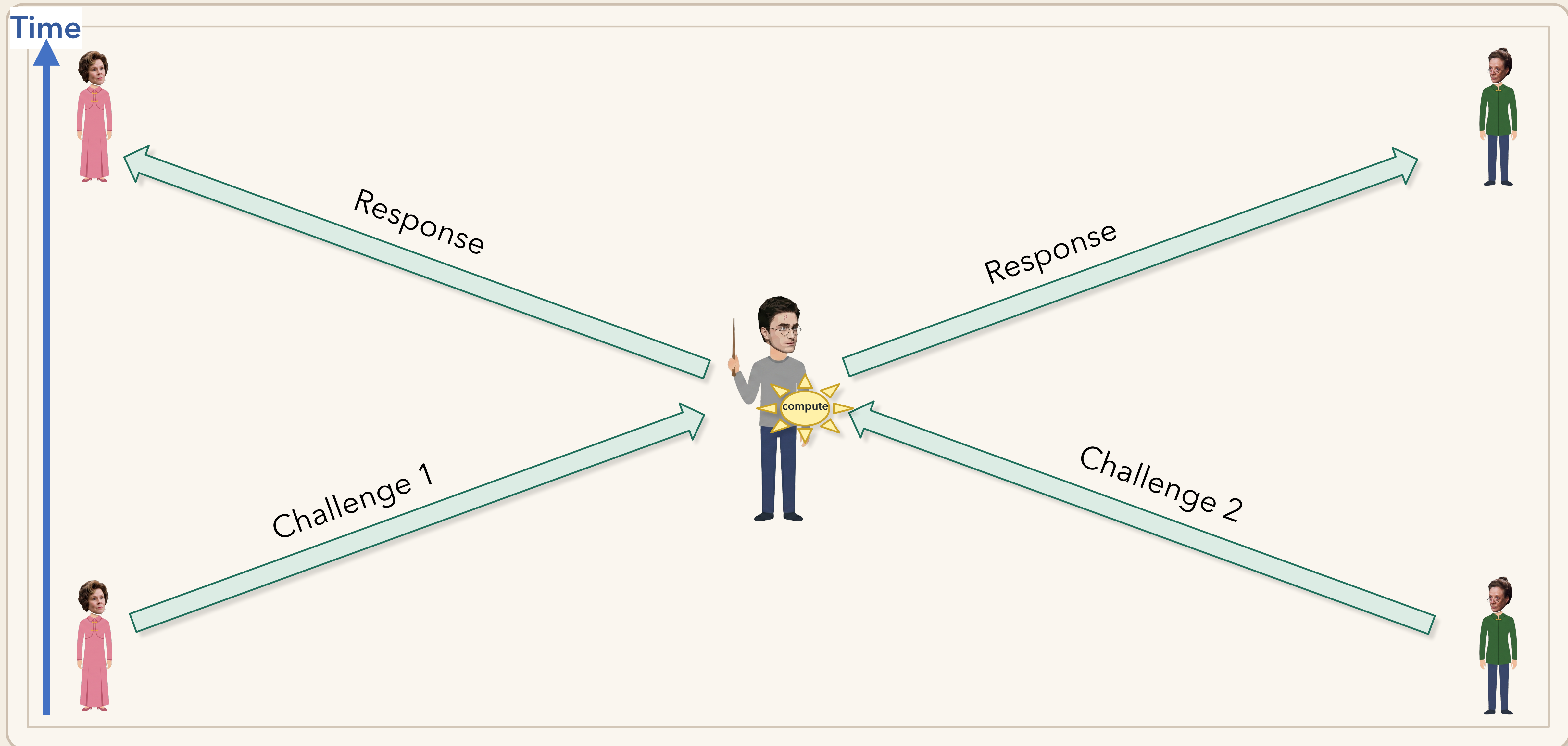
Visualizing PV in Spacetime



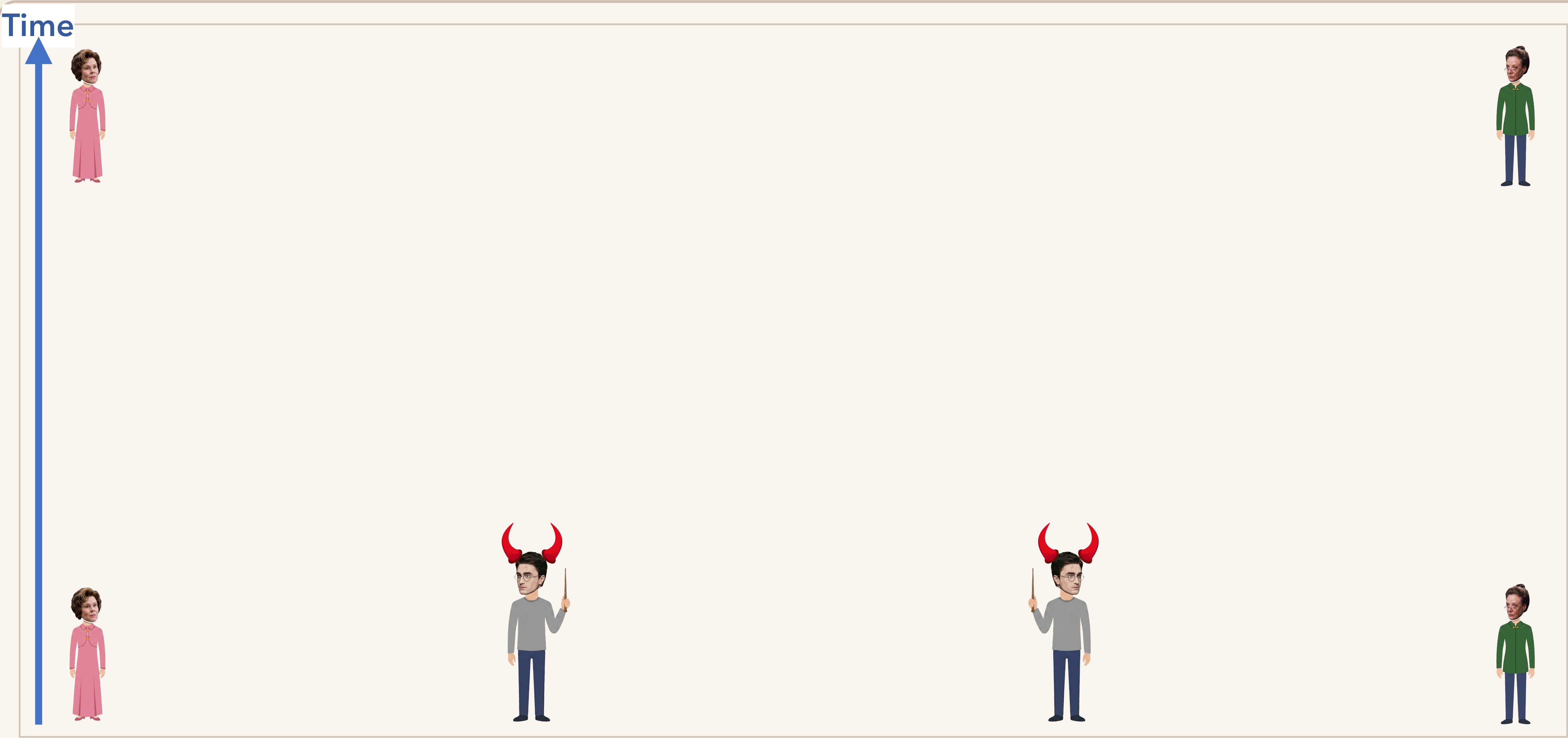
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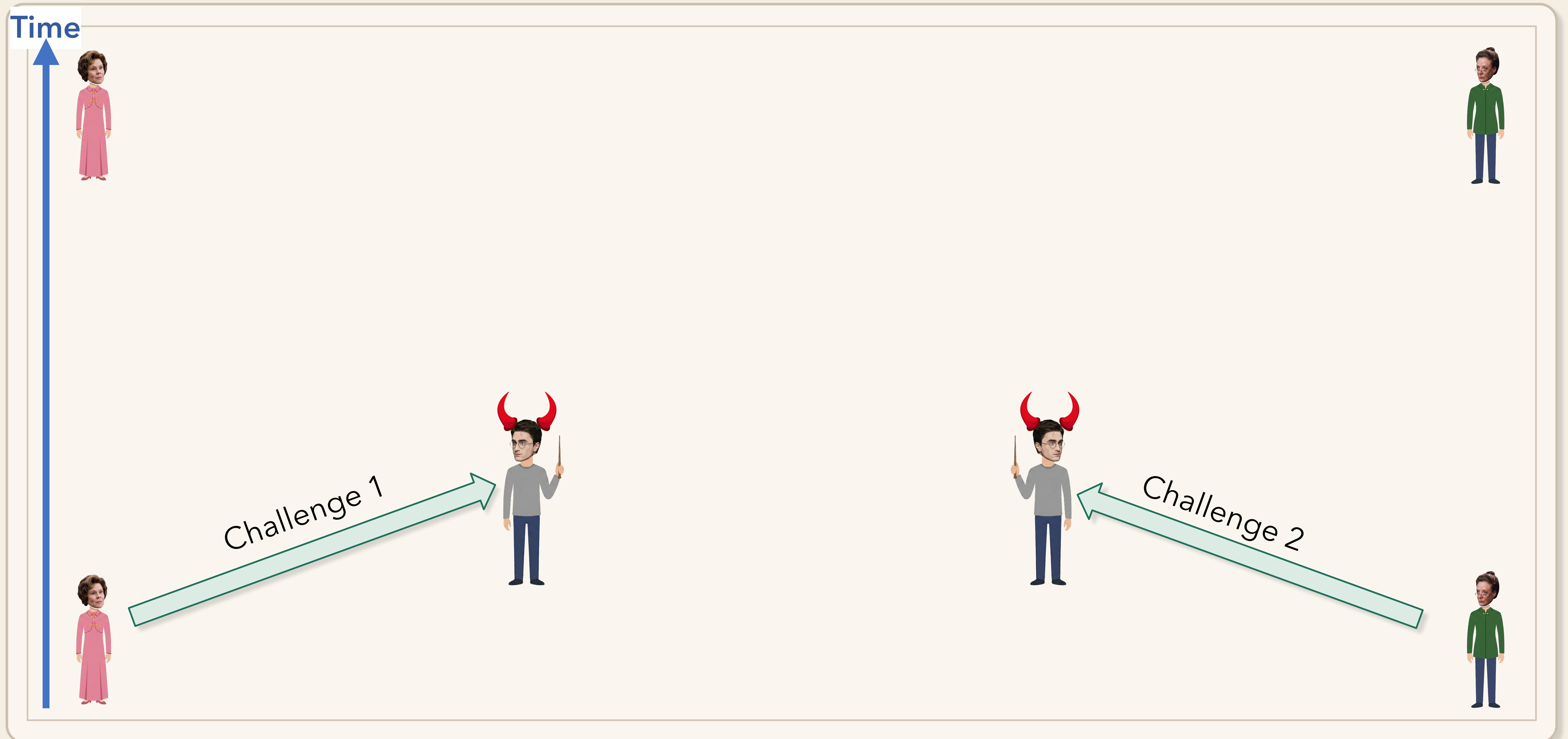
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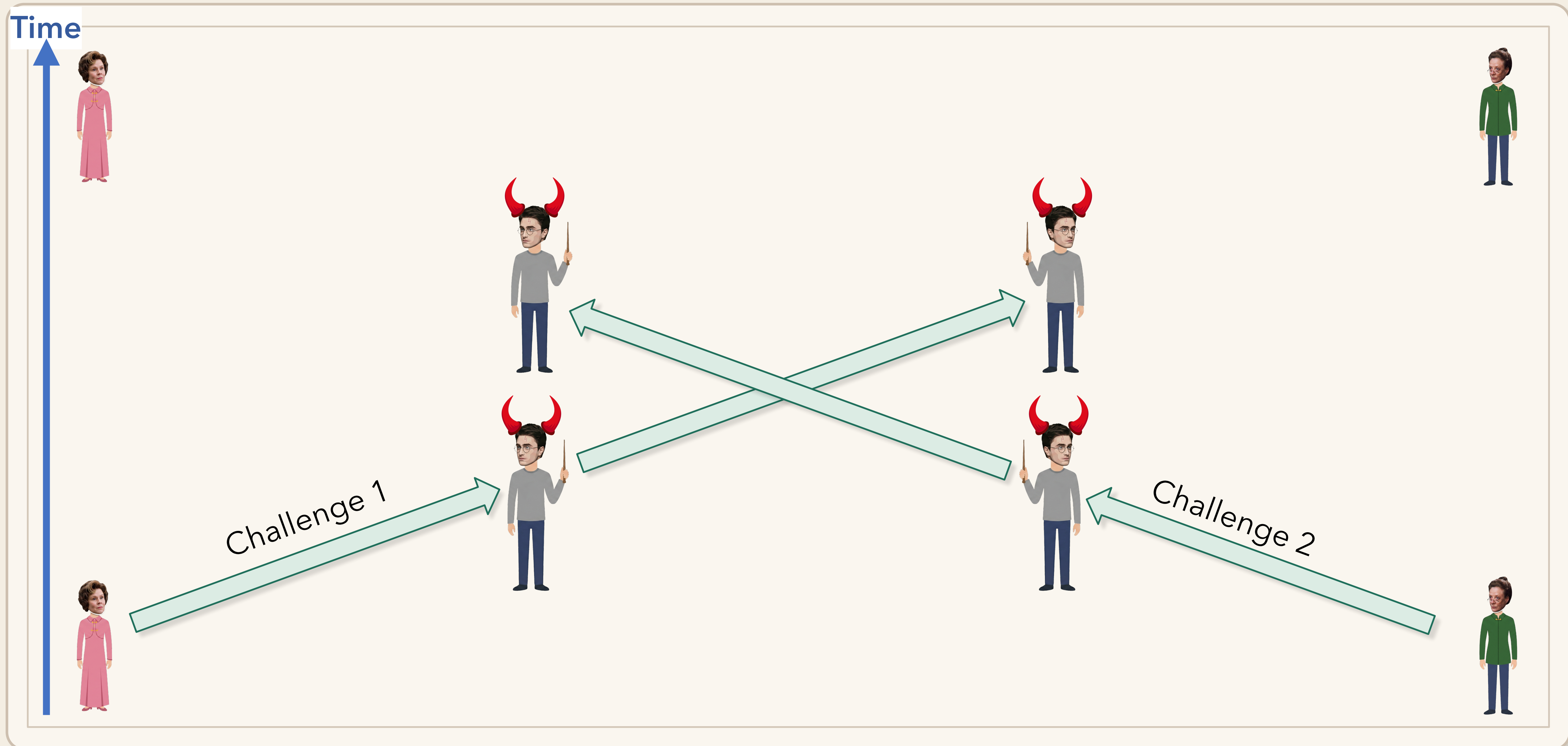
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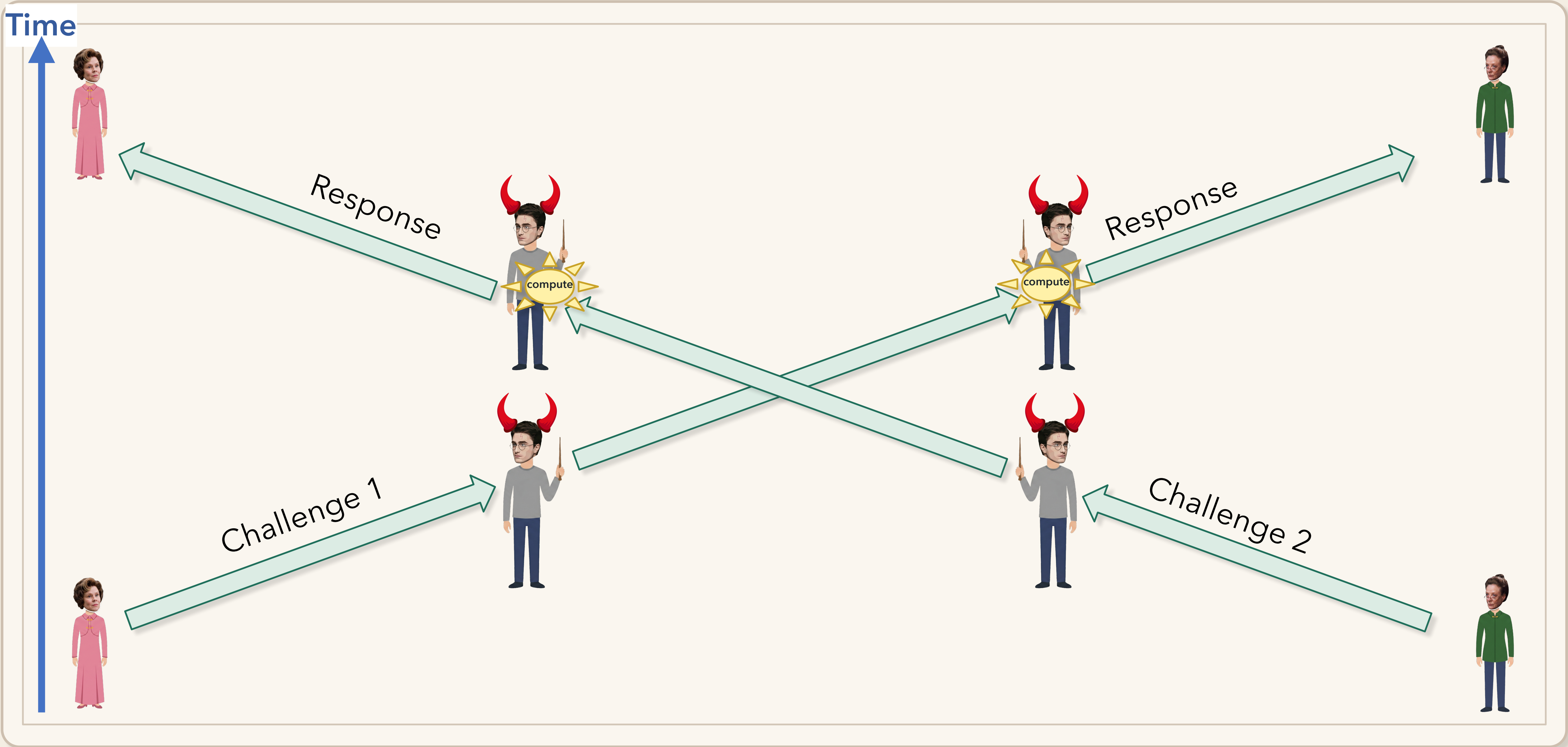
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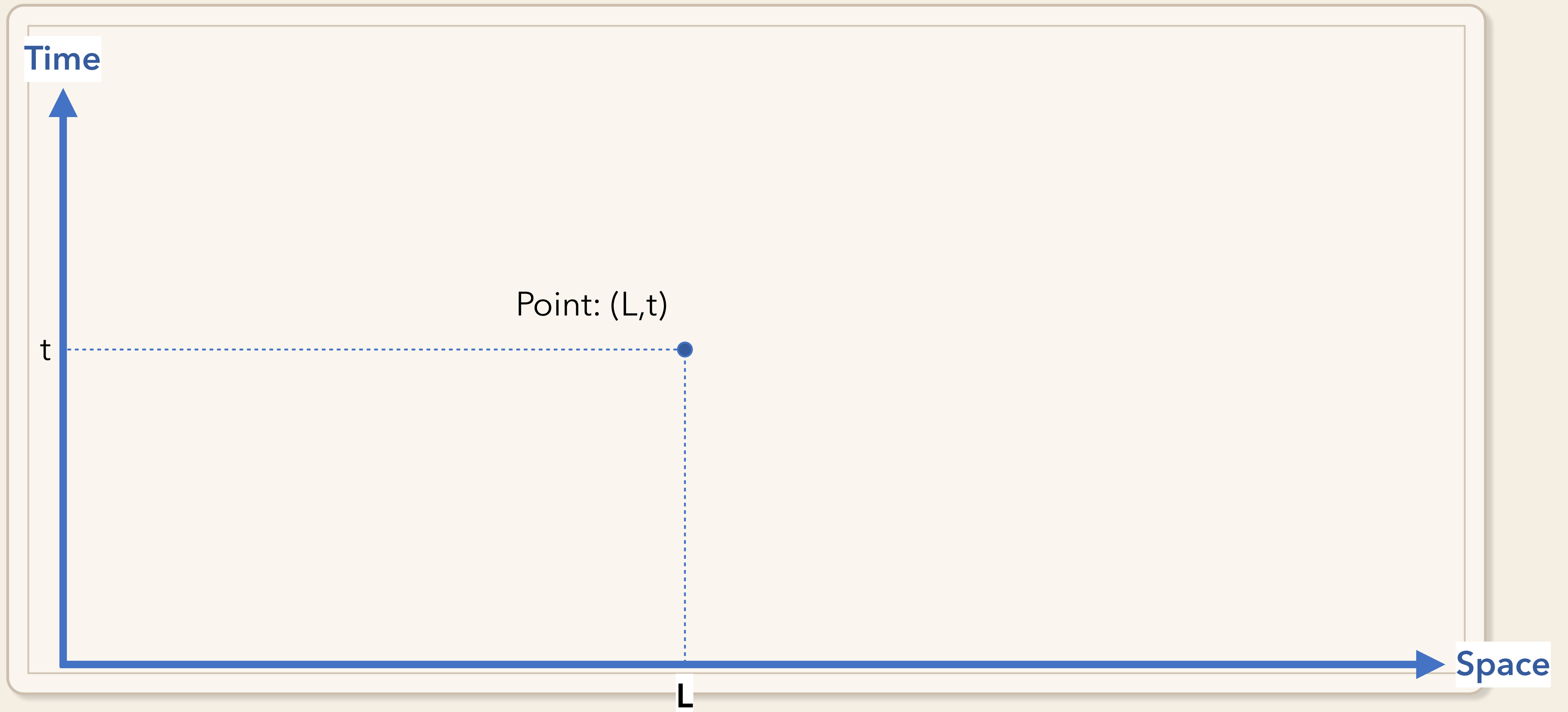


Our contributions:

1. Define ZK-PV: a **generalized** notion of position verification, **with privacy**
2. Construct ZK-PV from PV

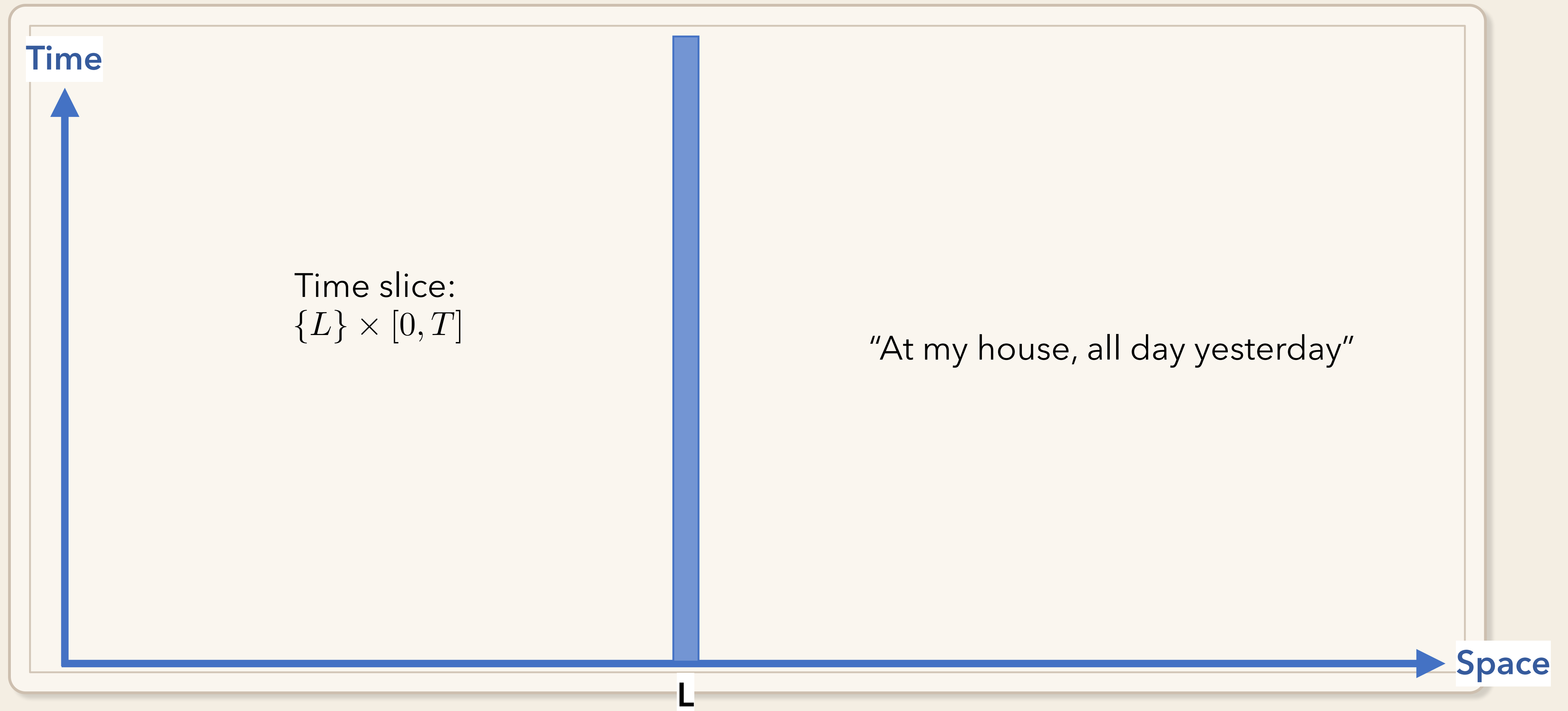
Proofs of **When** and **Where**

Generalize from proofs about exact position, to proofs about a **region of spacetime**



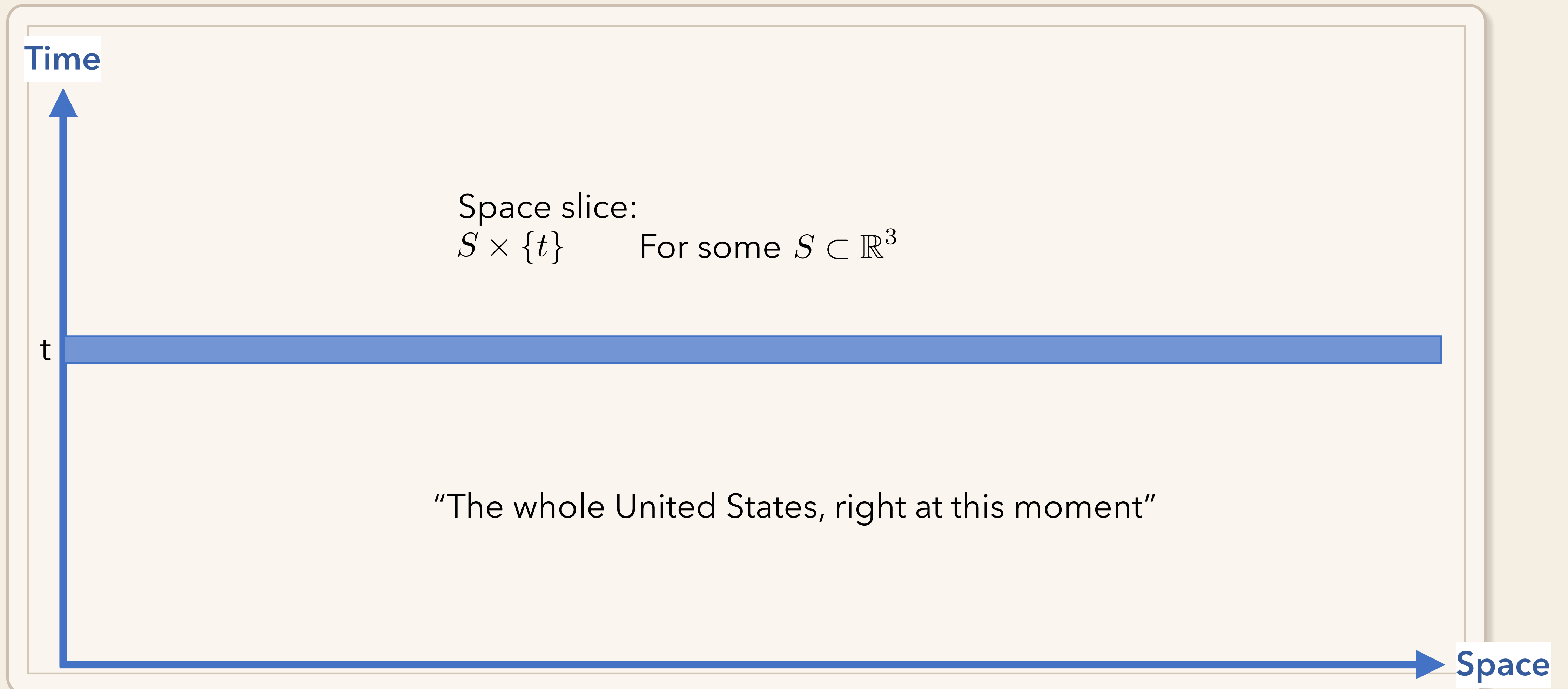
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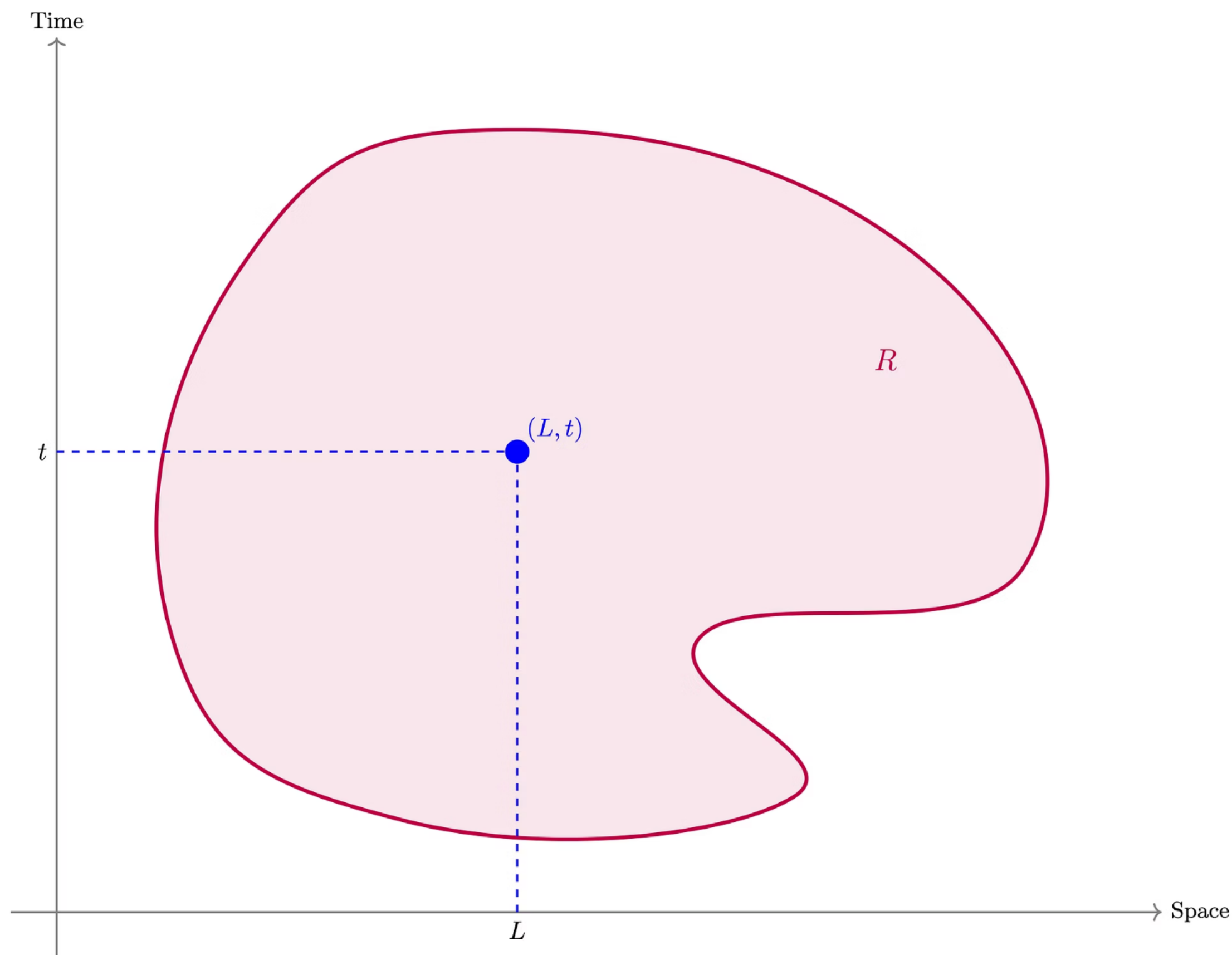
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Proofs of **When and Where**

Generalize from proofs about exact position, to proofs about a **region of spacetime**



Arbitrary region:

Can think of as corresponding to some Boolean predicate over spacetime coordinates (L, t)

Not necessarily contiguous.

Achieving Proofs of When and Where

How can we prove statements over arbitrary spacetime regions?

Achieving Proofs of When and Where

How can we prove statements over arbitrary spacetime regions?



Scenario 1

Achieving Proofs of When and Where

How can we prove statements over arbitrary spacetime regions?

Peter: Here, I'll prove to you that my elevation is really 350m



Scenario 1

Achieving Proofs of When and Where

How can we prove statements over arbitrary spacetime regions?



Scenario 1

Achieving Proofs of When and Where

How can we prove statements over arbitrary spacetime regions?

Accept ✓

Accept ✓



Scenario 1

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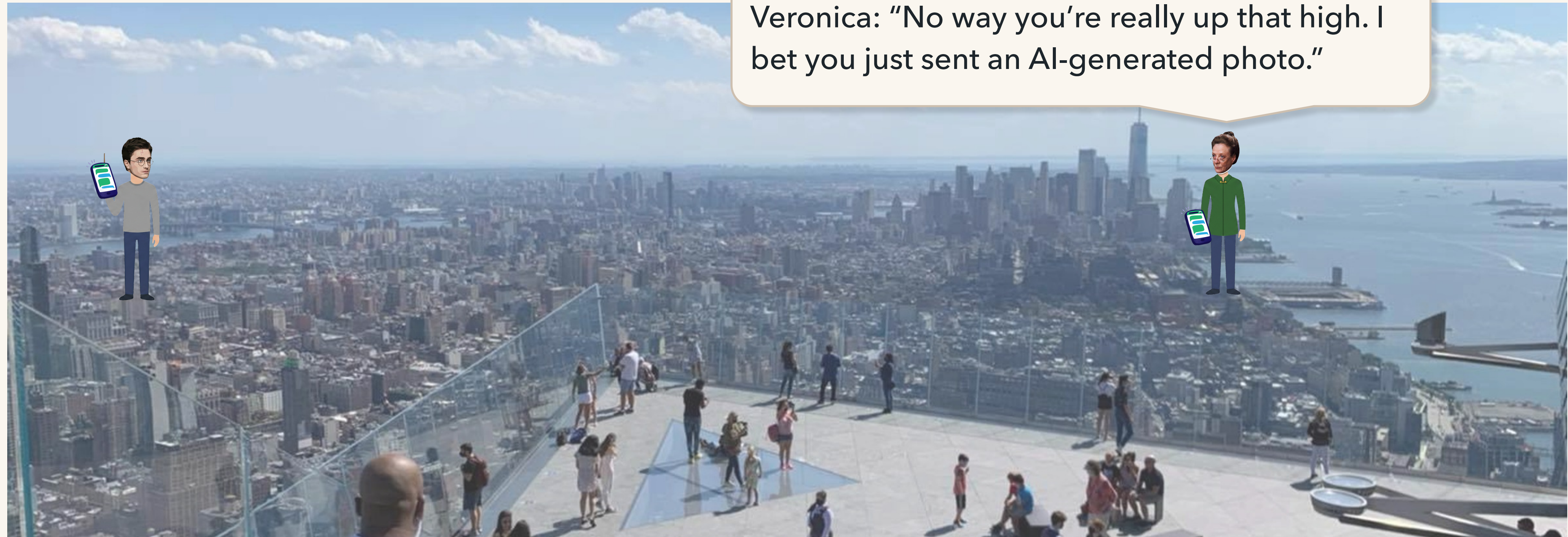
thinking If only there was a way to *just* prove that my location has a certain property (e.g. elevation), but keep the rest hidden...



Scenario 1

Achieving Proofs of When and Where

How can we prove statements over arbitrary spacetime regions?

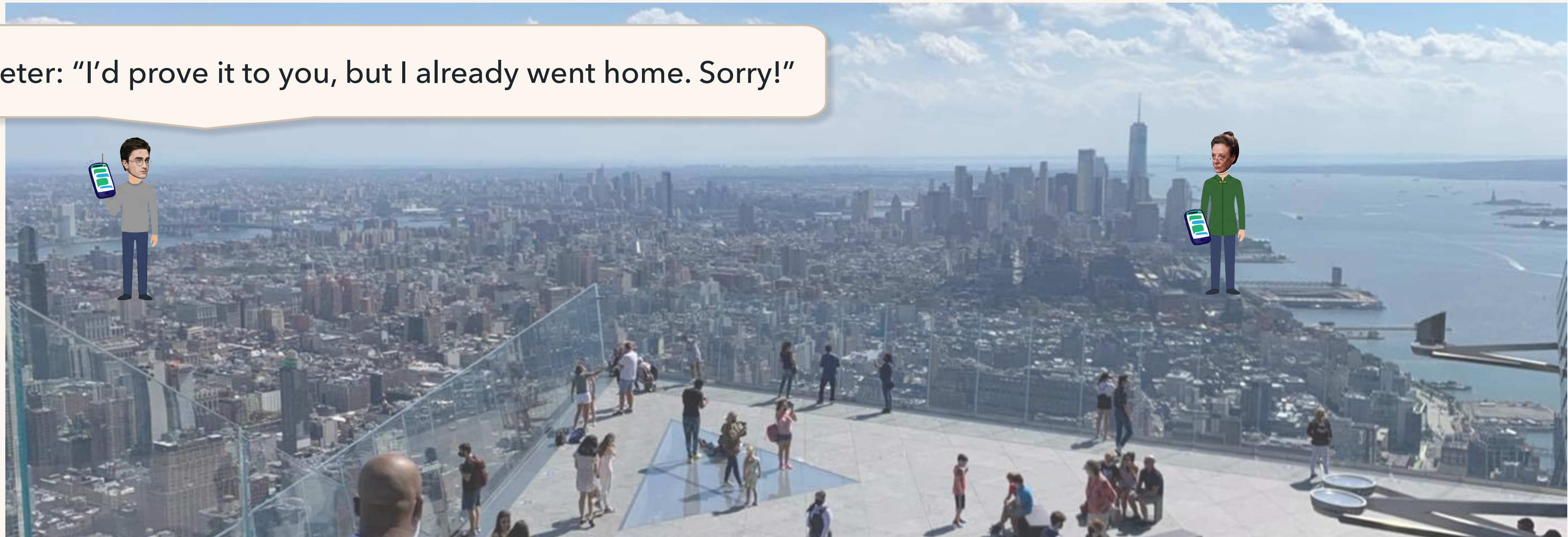


Scenario 2

Achieving Proofs of When and Where

How can we prove statements over arbitrary spacetime regions?

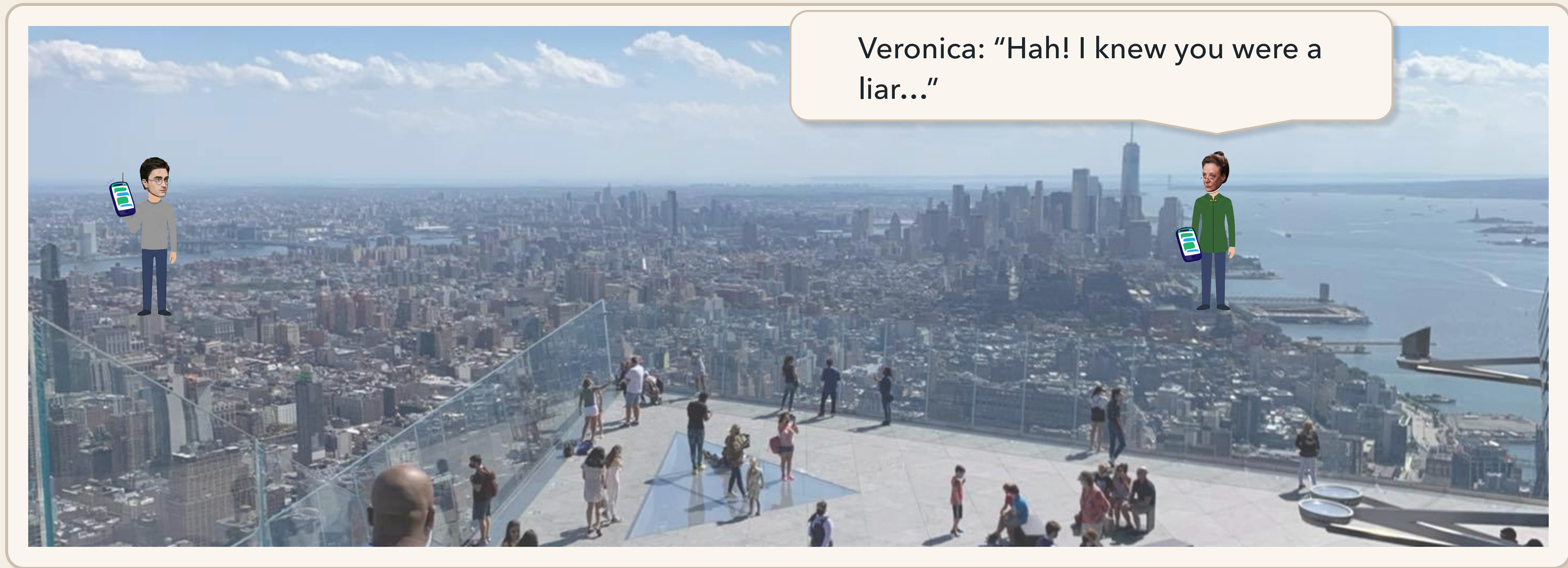
Peter: "I'd prove it to you, but I already went home. Sorry!"



Scenario 2

Achieving Proofs of When and Where

How can we prove statements over arbitrary spacetime regions?



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Achieving Proofs of When and Where

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Scenario 2

Achieving Proofs of When and Where

How can we prove statements over arbitrary spacetime regions?

Peter: *thinking* Am I just supposed to record my location all the time? That's a total invasion of privacy!



Scenario 2

Privacy is Non-Negotiable

To use “proofs of when and where” in real applications, we *need* to put privacy first.

Treaty Enforcement

*“No nation-state shall have a nuclear arsenal on Antarctica”
(Antarctic Treaty)*

Idea: require nations to *prove* that all of their weapons are outside of Antarctica.

Clearly, true locations must stay hidden.

Personal Privacy

Location leaks could have serious consequences. Cannot force users to constantly run PV.

Want private ways to prove:

- Alibis
- Device georestrictions
- Strava activities

Strava Security— Seriously!



FRANCE • STRAVALEAKS

**How Emmanuel Macron
can be tracked: Watch
the first episode of
StravaLeaks**

Application: proving properties of our spacetime *trajectory* – such as how far we ran, or how quickly, or in what shape – while preserving privacy.

Our Solution: Zero-Knowledge Position Verification

We define a new primitive, to be the basis for *private* position-based cryptography

Take a spacetime region R . A **ZK-PV protocol** for R should satisfy three properties:

- 1 Completeness: A prover anywhere in R is accepted by the protocol.
- 2 Soundness: A prover located *outside* of R is rejected w.h.p. by the protocol.
- 3 Zero-Knowledge: The verifiers learn *nothing* about the prover's location, besides the fact that it is in R .

Main Result: Compiling PV to ZK-PV

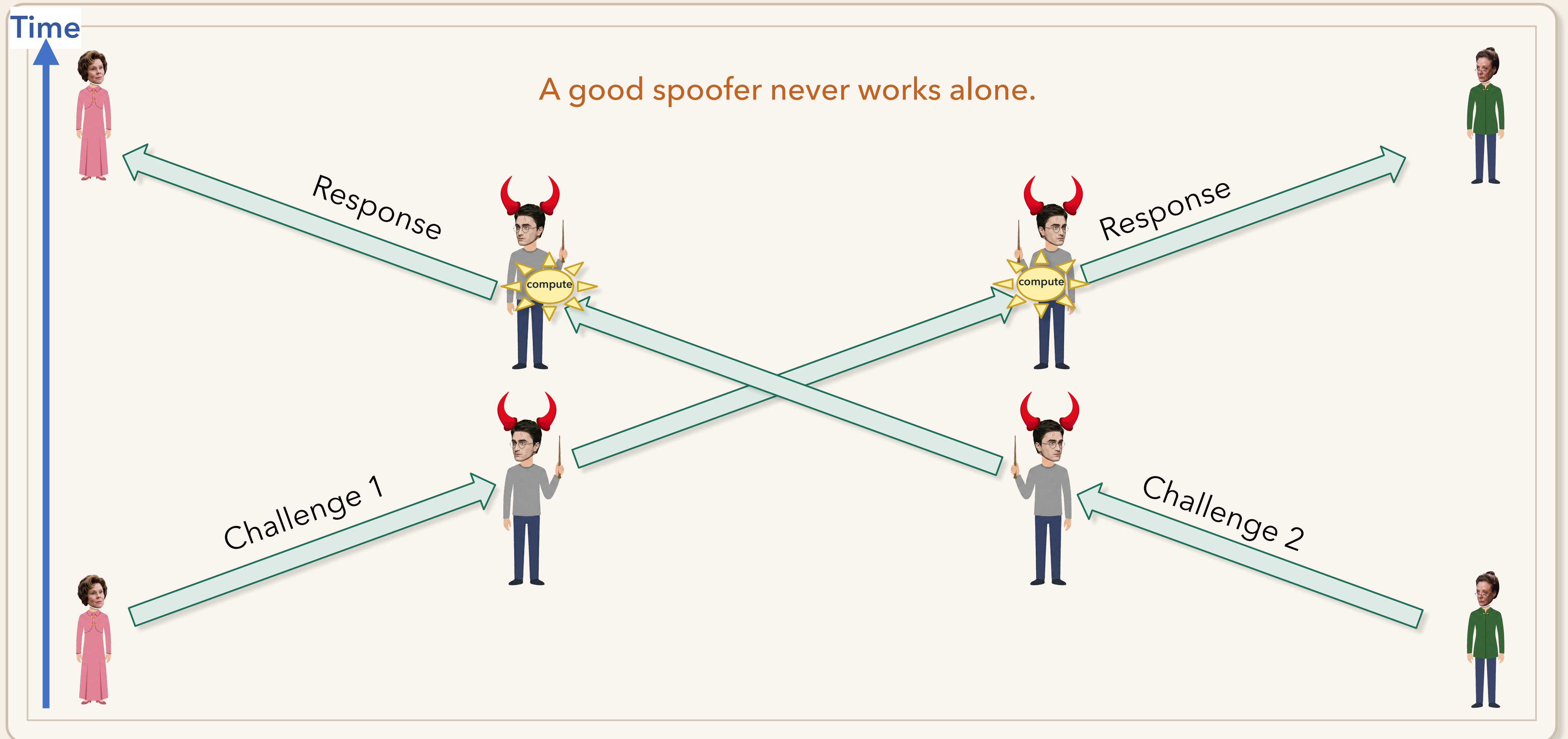
Assuming post-quantum secure one-way functions, any* position verification protocol can be upgraded to a zero-knowledge** position verification protocol, through a *black-box* transformation.

*As long as it is single-round, with classical outputs

**Honest-verifier zero knowledge.
We'll talk more about this later.

Part Two: Our Construction

Position Verification Against Colluding Adversaries



Classical Impossibility

Theorem (CGMO'09)

Every classical position verification scheme is susceptible to an efficient attack by colluding spoofers.

Intuition

The generic attack is an extension of the 1D case:

Copy the verifier challenges and forward them along.

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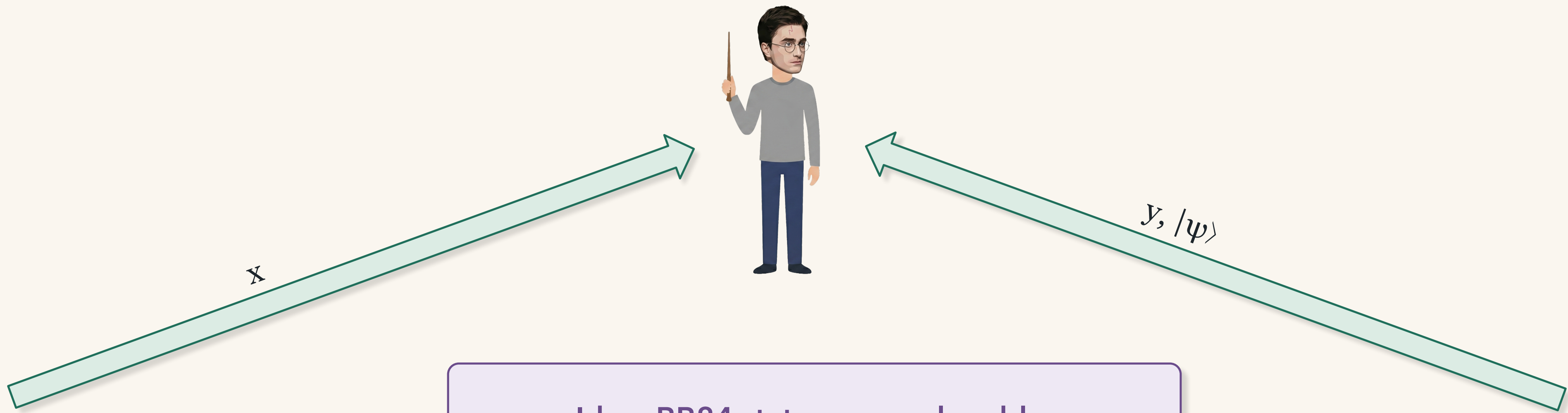
How to get around this? [Quantum!](#)

Quantum Position Verification: f-BB84 Scheme

Time

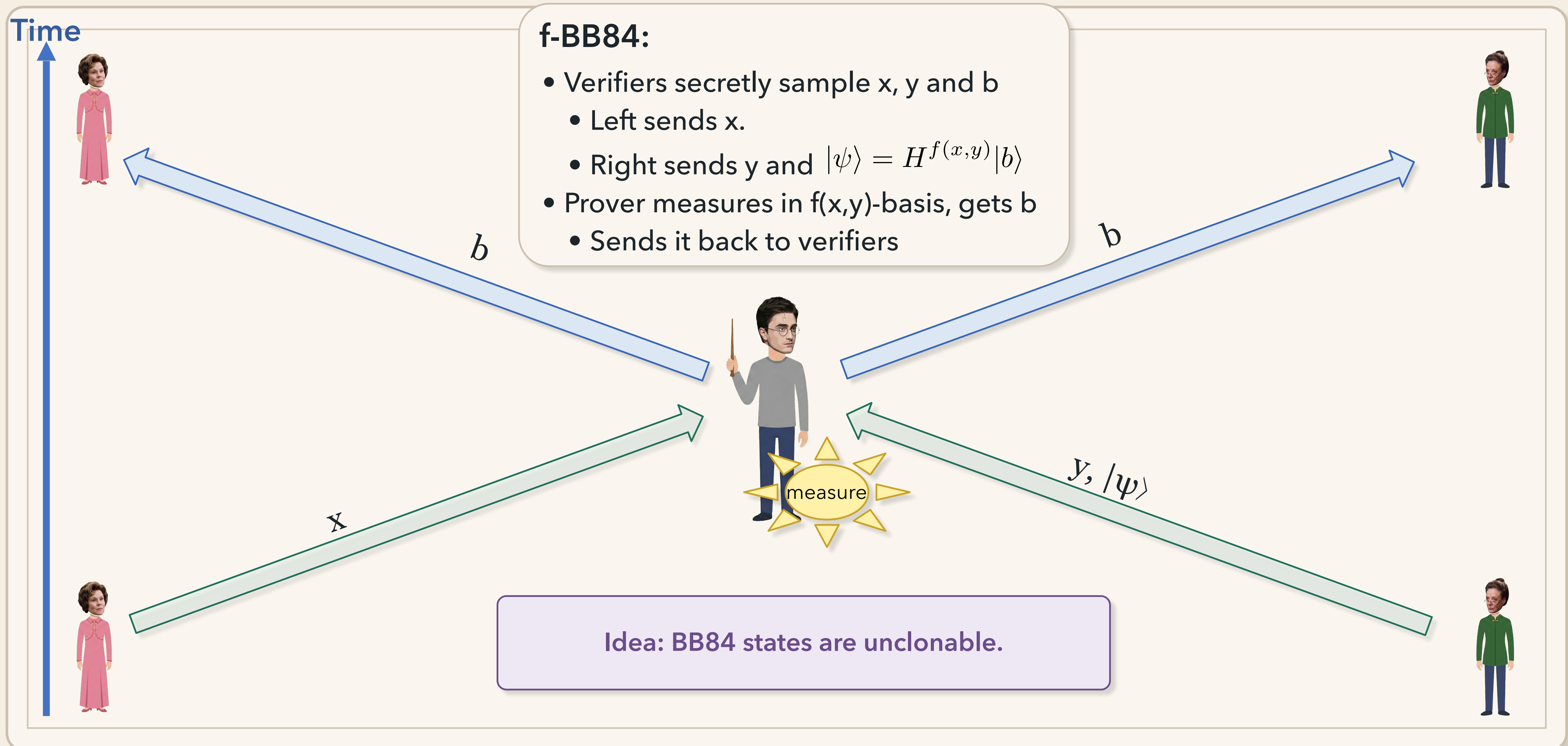
f-BB84:

- Verifiers secretly sample x, y and b
 - Left sends x .
 - Right sends y and $|\psi\rangle = H^{f(x,y)}|b\rangle$

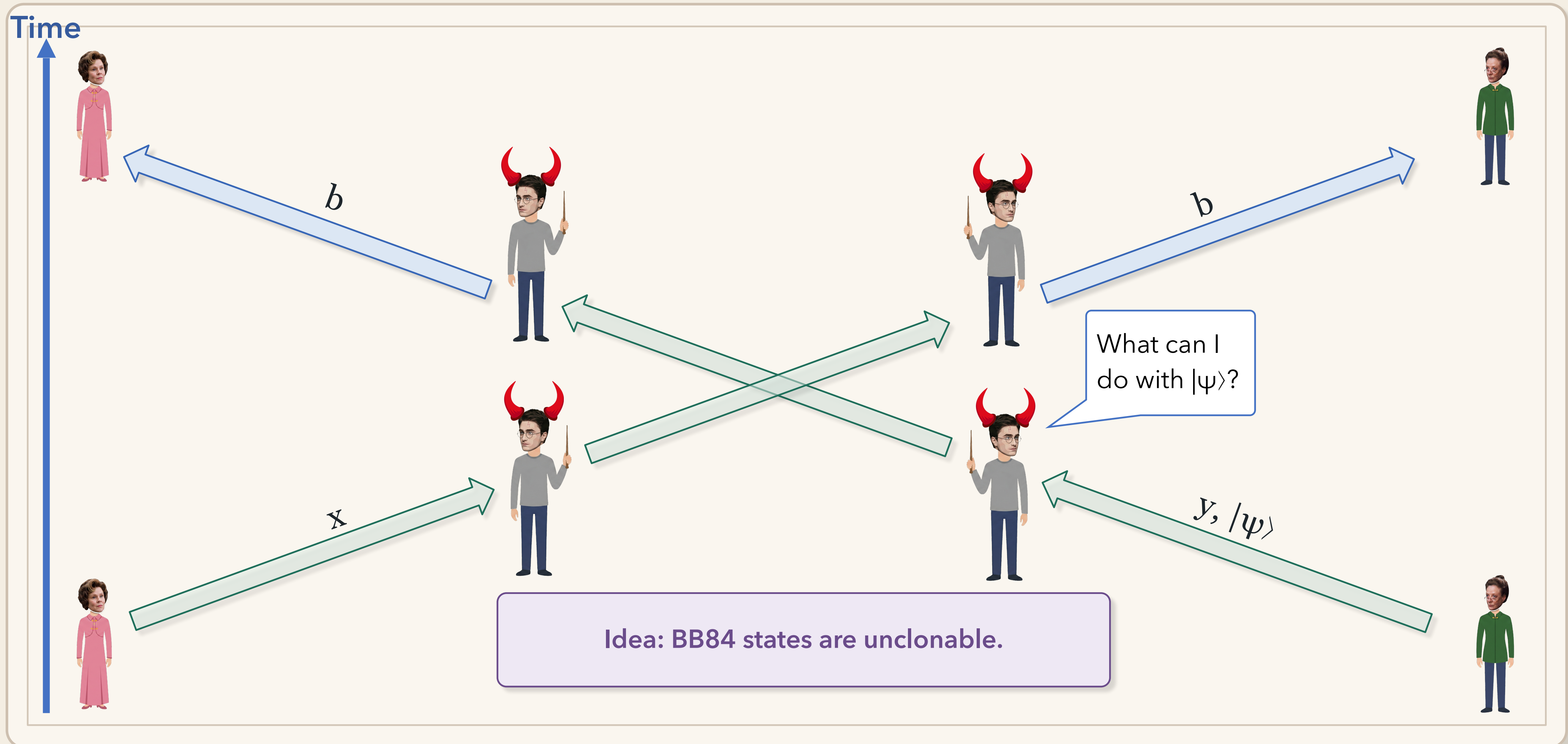


Idea: BB84 states are unclonable.

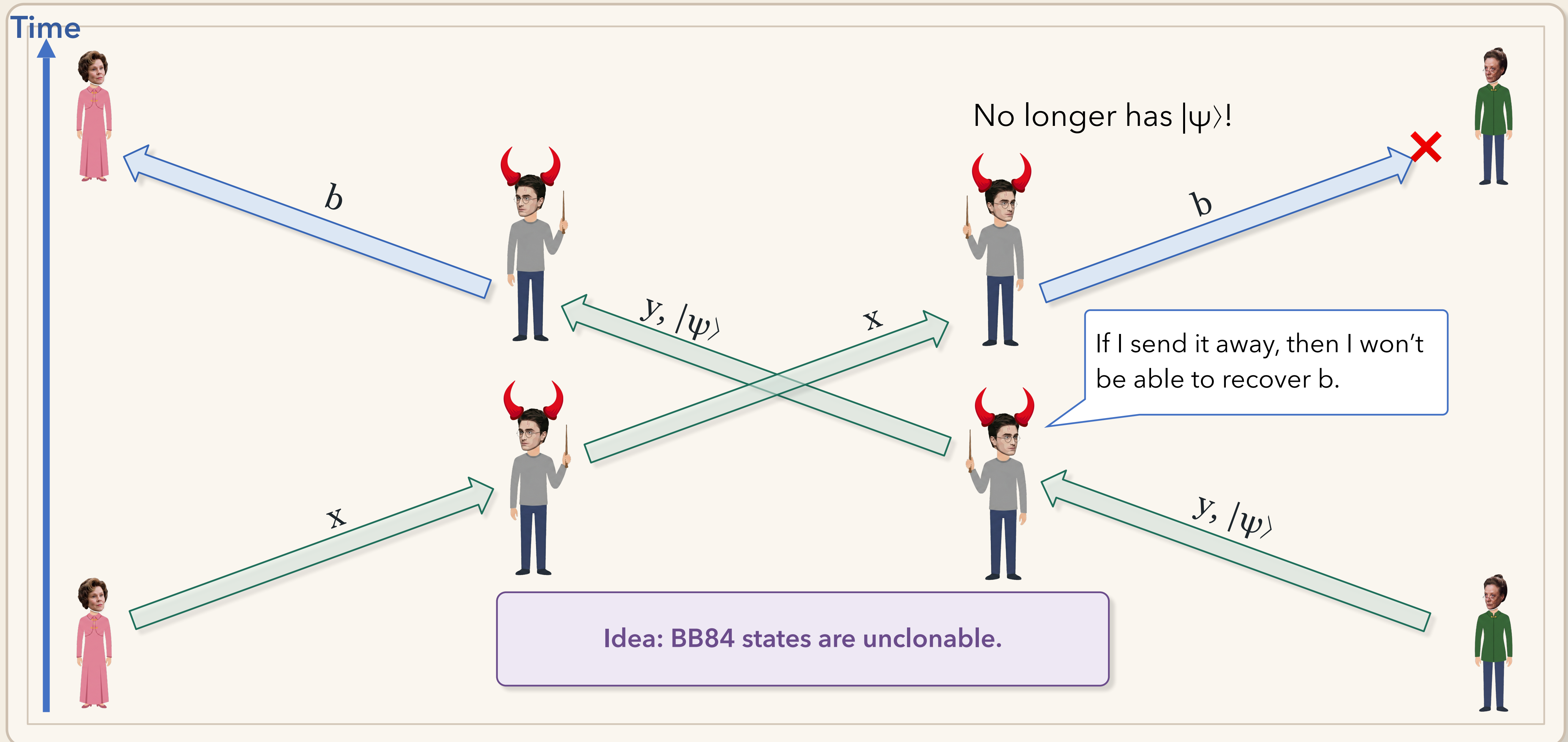
Quantum Position Verification: f-BB84 Scheme



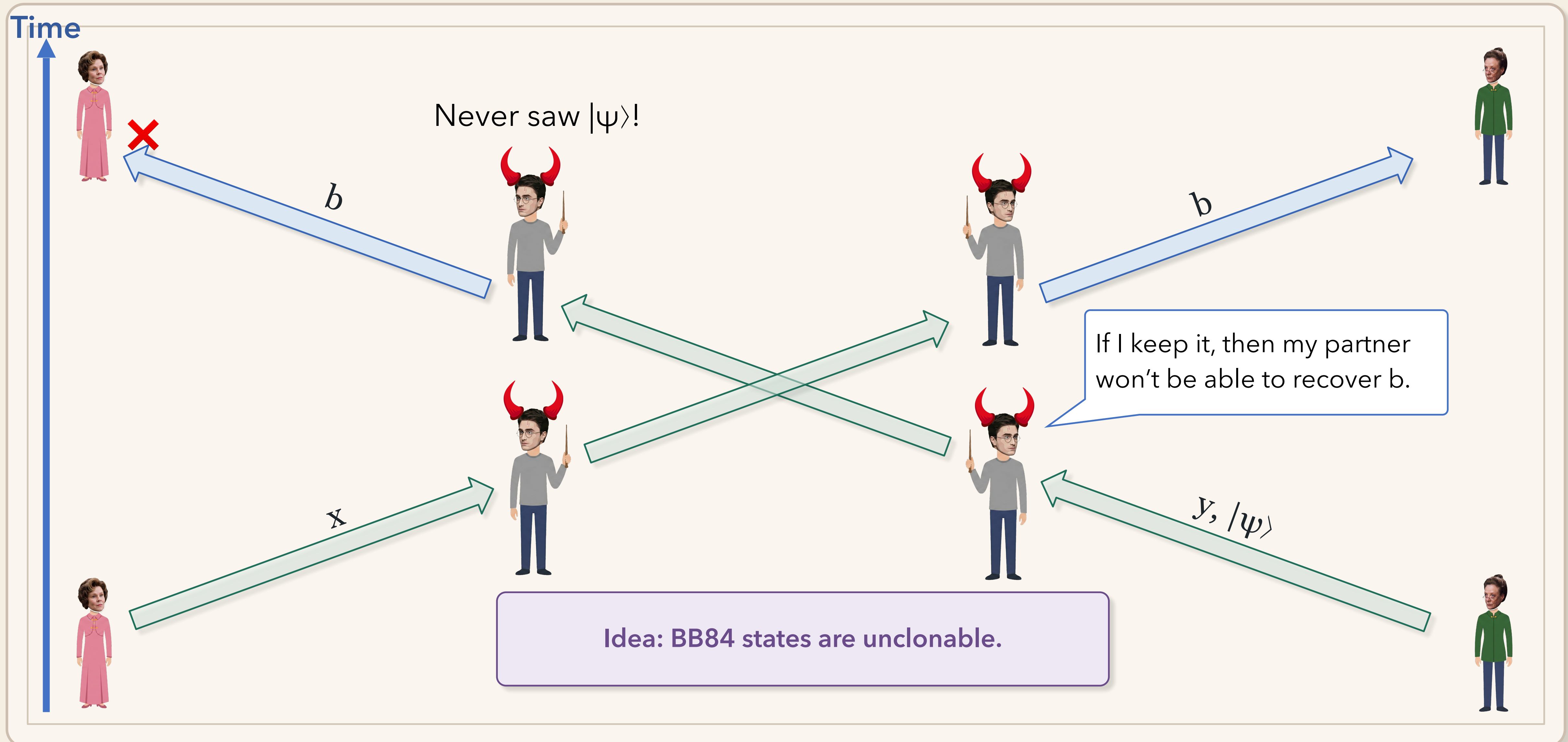
Quantum Position Verification: f-BB84 Scheme



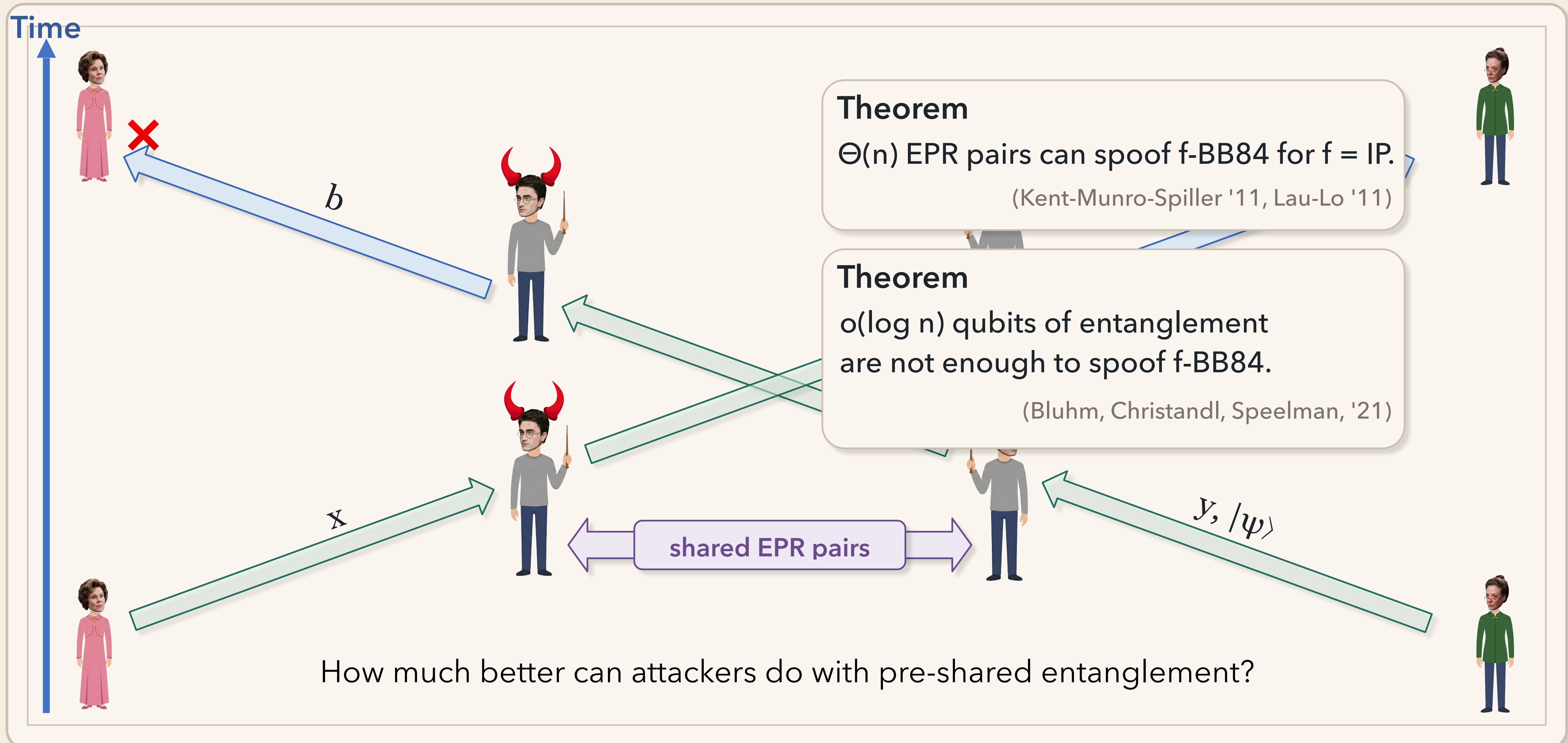
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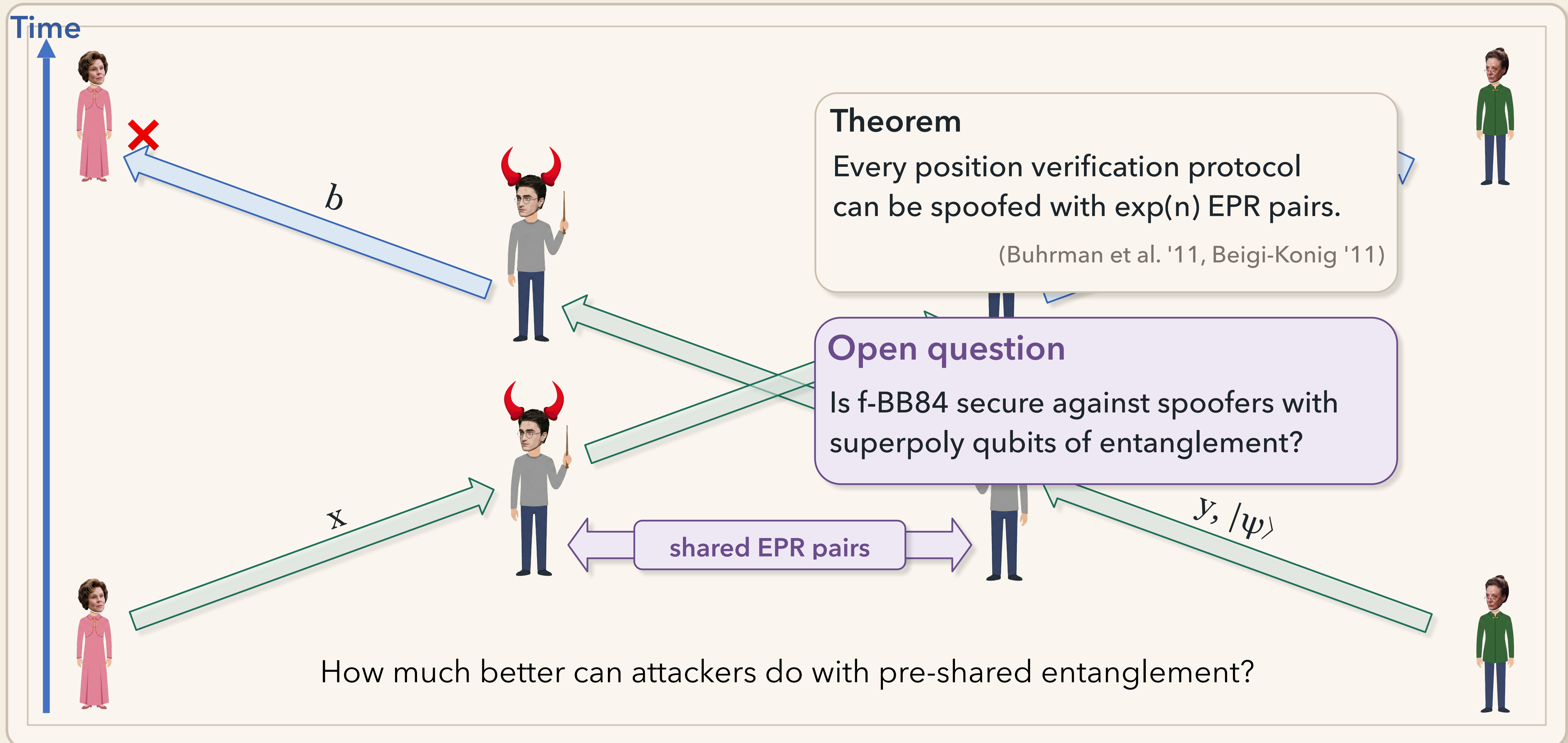
Quantum Position Verification: f-BB84 Scheme



Quantum Position Verification: f-BB84 Scheme



Quantum Position Verification: f-BB84 Scheme



QPV — Not Just a Theoretical Pipe Dream!

Quantum Position Verification with Remote Untrusted Devices

Gautam A. Kavuri,^{1,2,*} Yanbao Zhang,^{3,†} Abigail R. Gookin,^{4,2} Soumyadip Patra,⁵ Joshua C. Bienfang,⁶ Honghao Fu,⁷ Yusuf Alnawakhtha,⁸ Dileep V. Reddy,^{1,2} Michael D. Mazurek,^{1,2} Carlos Abellán,⁹ Waldimar Amaya,⁹ Morgan W. Mitchell,^{10,11} Sae Woo Nam,^{1,12} Carl A. Miller,^{13,8} Richard P. Mirin,¹² Martin J. Stevens,¹² Scott Glancy,^{14,1} Emanuel Knill,^{14,1,15} and Lynden K. Shalm^{1,12,16,‡}

¹Department of Physics, University of Colorado, Boulder, CO, 80309, USA

²Associate of the National Institute of Standards and Technology, Boulder, CO, 80305, USA

³Quantum Information Science Section,
Computational Sciences and Engineering Division,
Oak Ridge National Laboratory, Oak Ridge, TN, 37831, USA

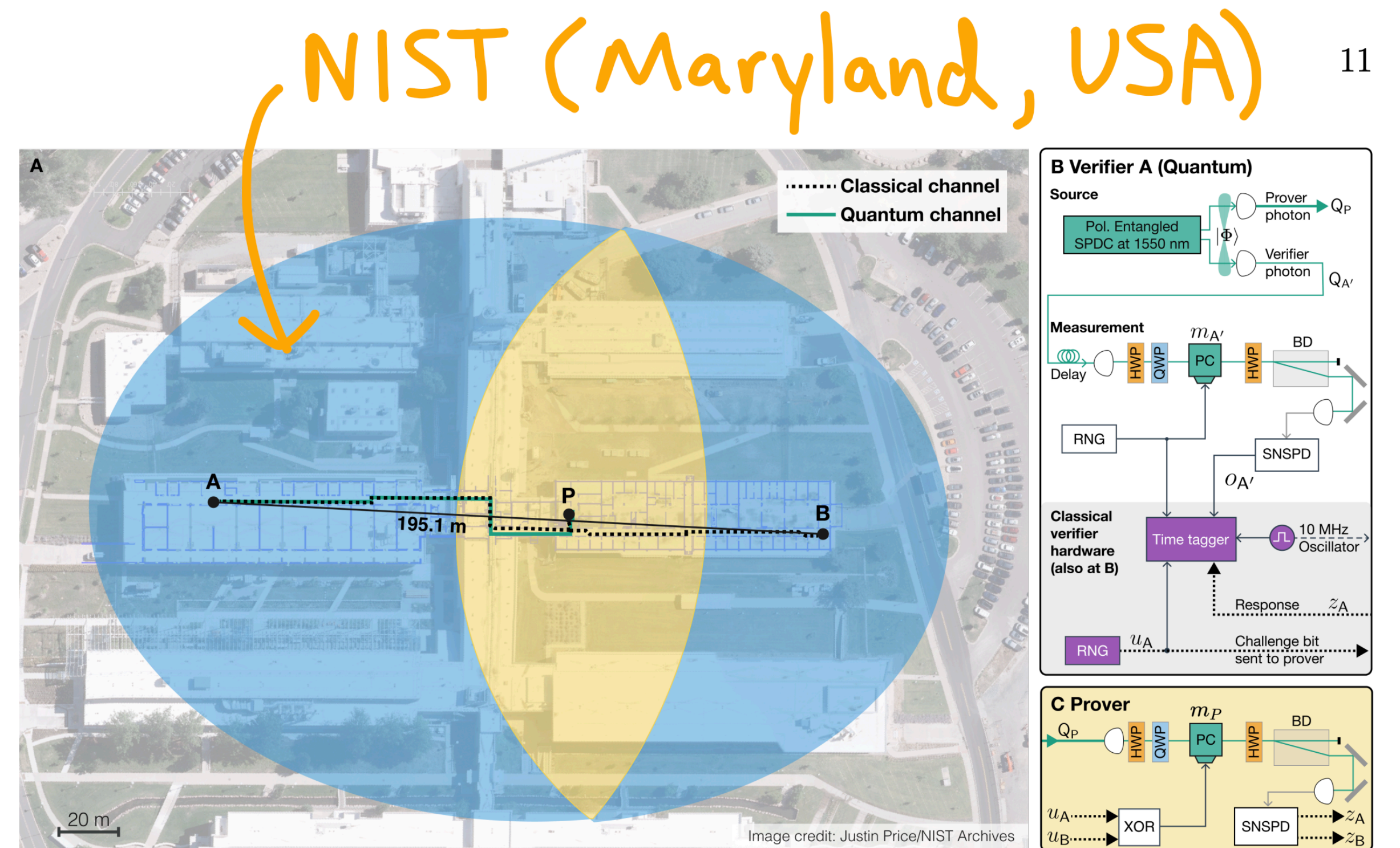


FIG. 3. Figure depicting the experimental setup and verifier and prover stations used for our demonstration. Subfigure A depicts the quantum target region E of our protocol in yellow, and the classical target region of a comparable classical protocol in blue overlaid on an aerial photograph of the NIST building where the experiment is housed. Subfigure B shows a schematic of the

Making (Q)PV Protocols Private

Constructing ZK-PV: A Roadmap

The construction will proceed in two phases.

1 **Black-Box Compiler from PV to Position Commitments**

- Define position commitments – analogous to standard commitments.
- Construct a position commitment from any “nice” PV protocol.

Constructing ZK-PV: A Roadmap

The construction will proceed in two phases.

1 **Black-Box Compiler from PV to Position Commitments**

- Define position commitments – analogous to standard commitments.
- Construct a position commitment from any “nice” PV protocol.

2 **Combine Position Commitments with Classical ZK**

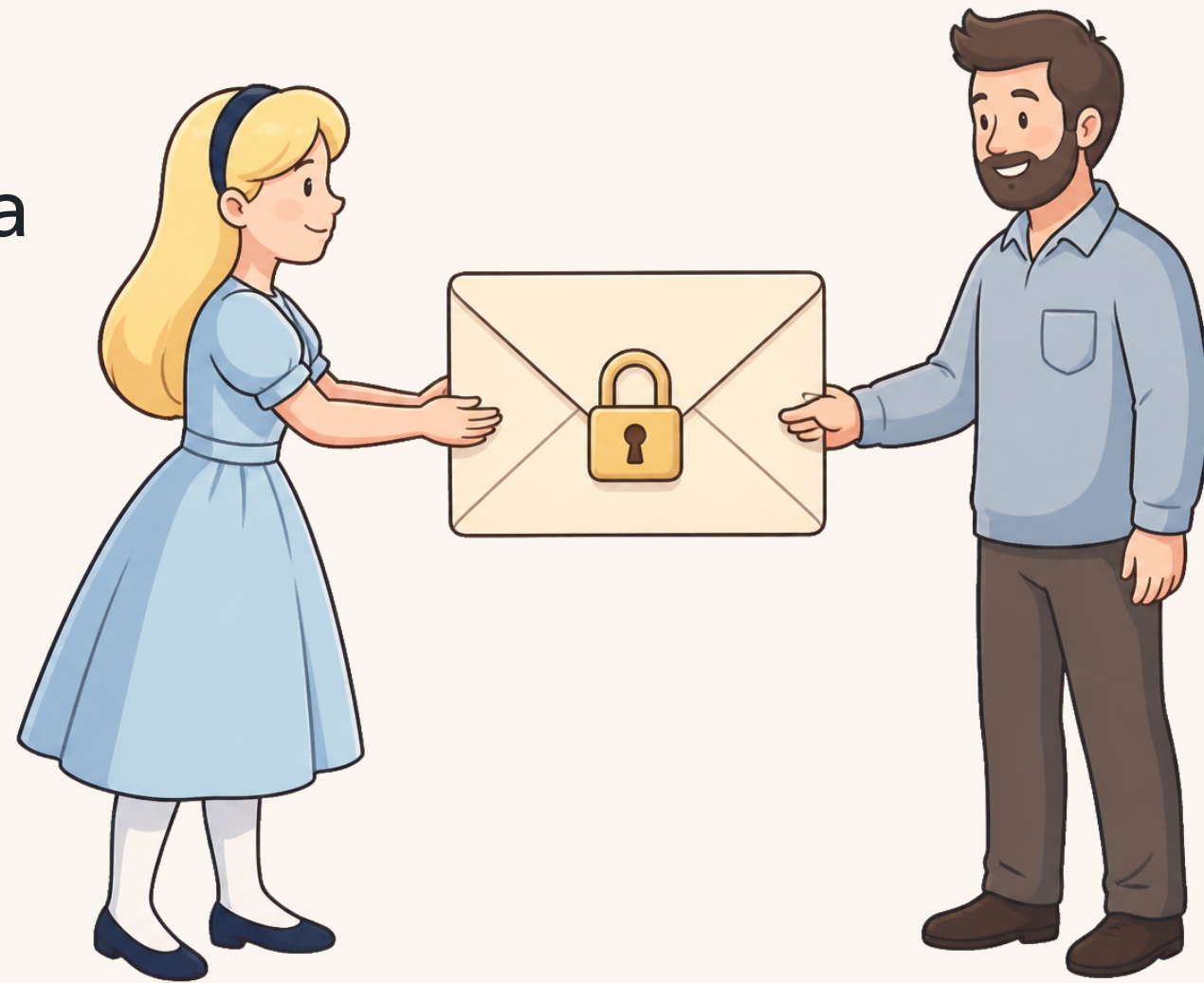
- Classical result: commitments $\rightarrow$ zero-knowledge proofs for NP
- Our setting: position commitments $\rightarrow$ zero-knowledge position verification

Refresher: Cryptographic Commitments

Commit

Alice sends Bob a message of her choice,

"Hidden in a padlocked envelope".



Reveal

She sends Bob a secret reveal string which opens the commitment.

He checks that it fits, and unlocks the message.



Security Properties

Hiding

Bob learns nothing about the message until Alice chooses to reveal.

Binding

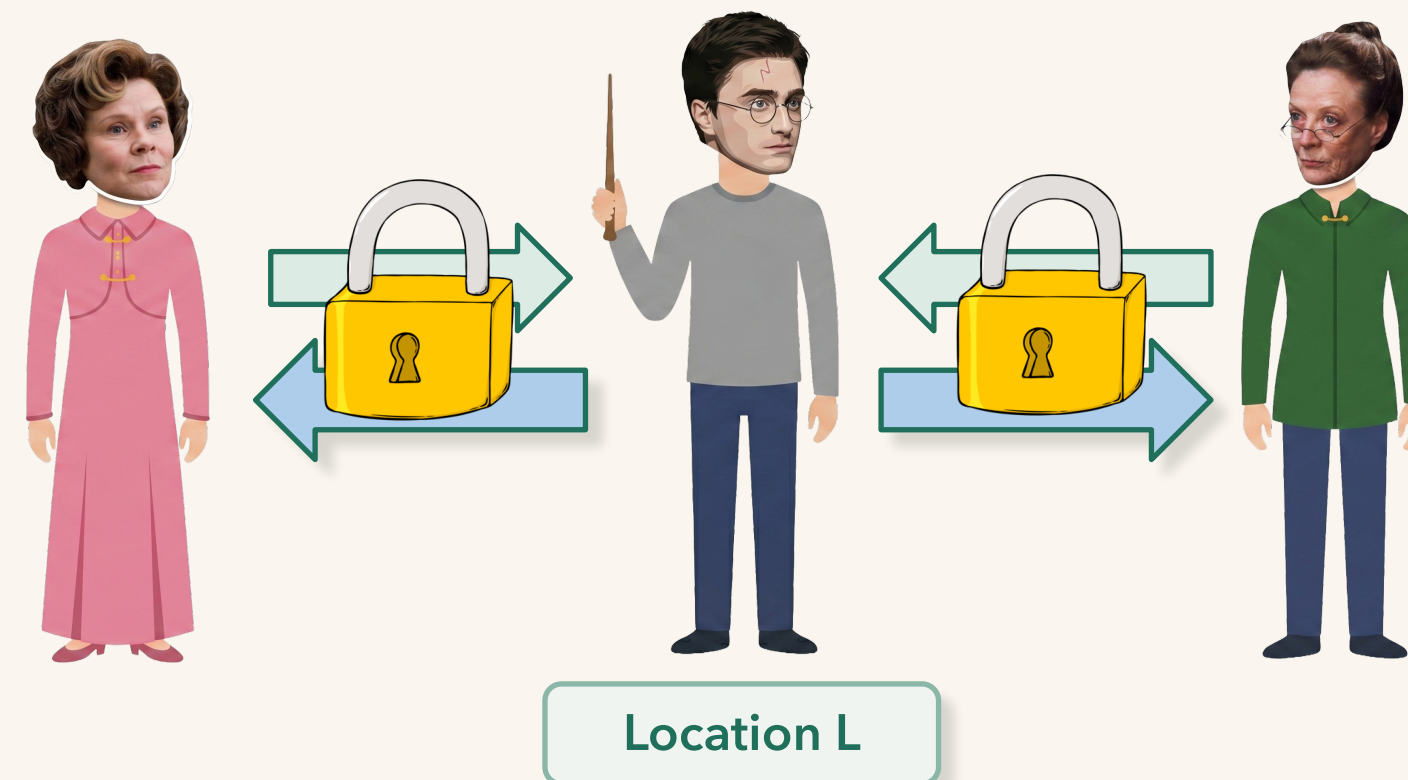
Once Alice commits, there is only one message she can successfully open to.

Position Commitments

Commit

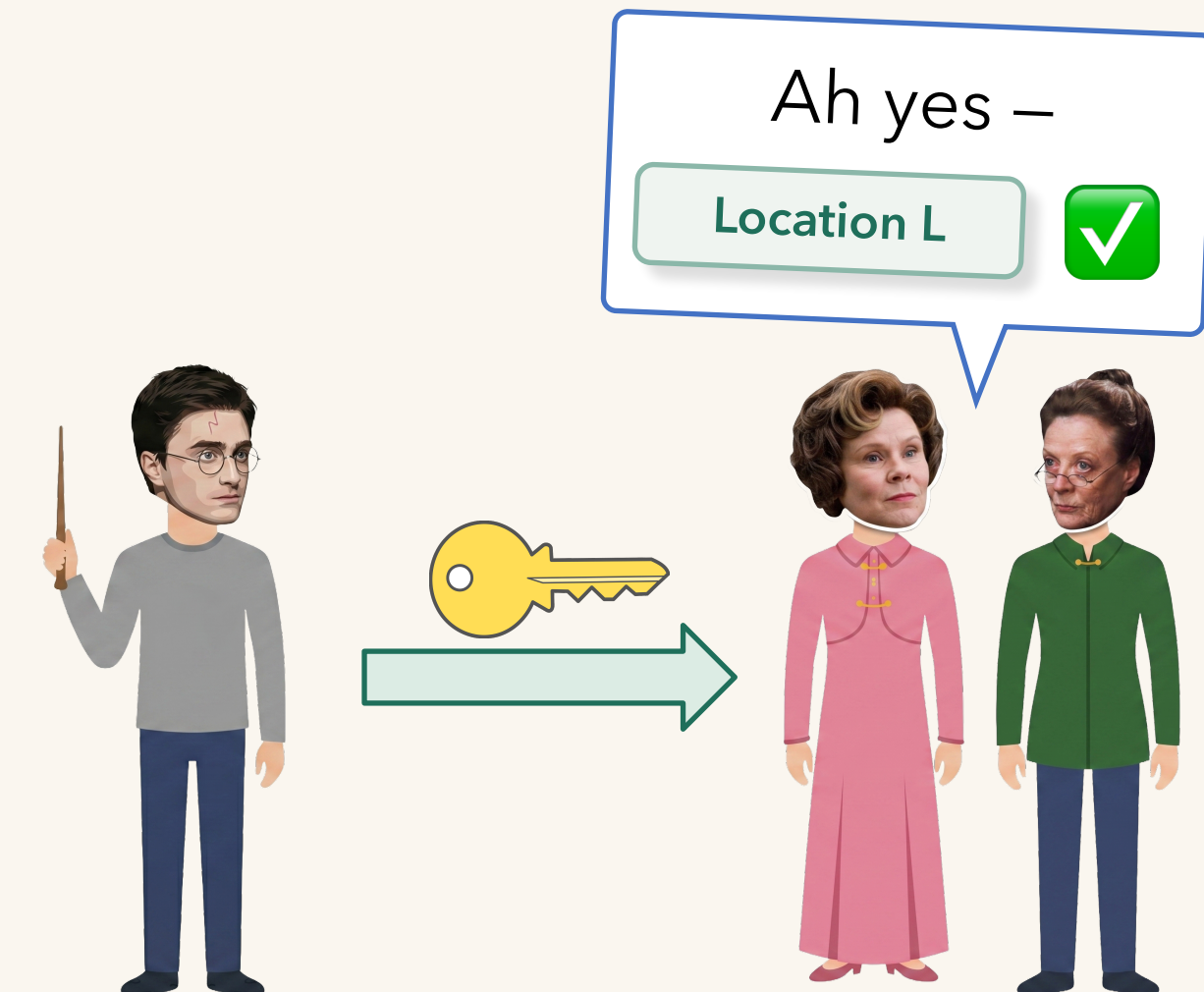
Peter commits to his true position,

But it is hidden from Veronica and Victoria.



Reveal

At a later time, Peter can choose to unseal his commitment by sending a classical reveal string.



Security Properties

Hiding

Before Peter sent a reveal, the verifiers' view tells them nothing about Peter's true location.

Position Binding

If Peter reveals to position (L,t) and verifiers accept, then he must have been located at (L,t) .

Black-Box Compiler: PV to Position Commitments

Main idea: run *parallel instances* of PV for different locations, and *commit to responses*



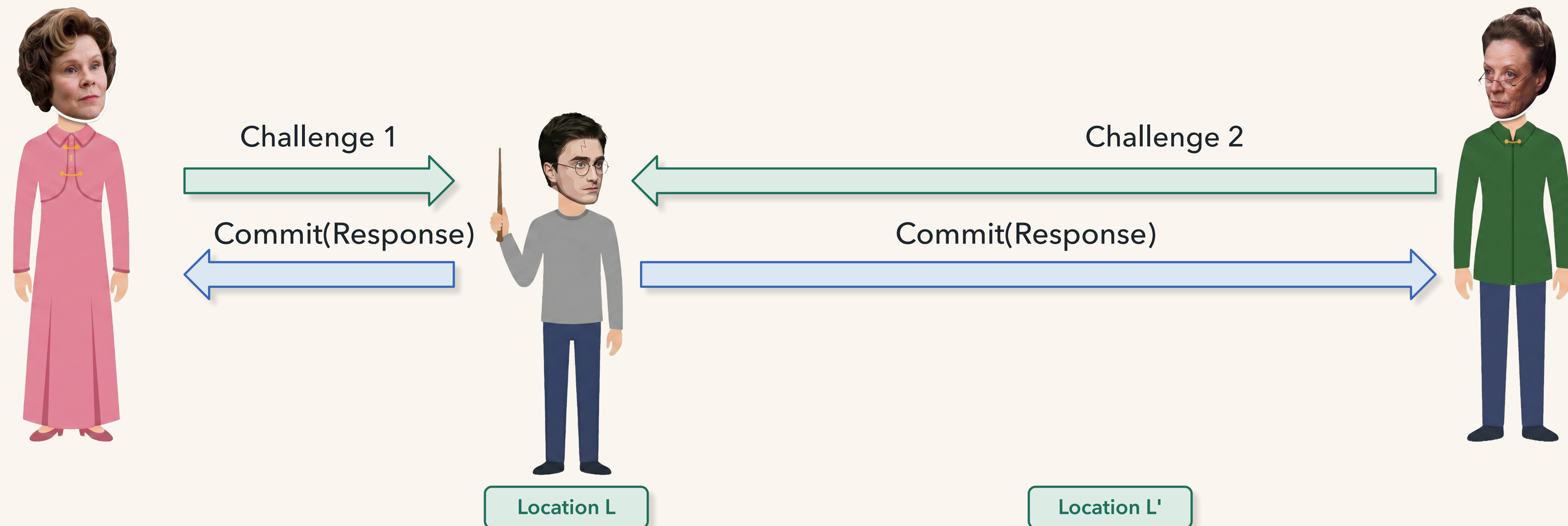
Location L



Location L'

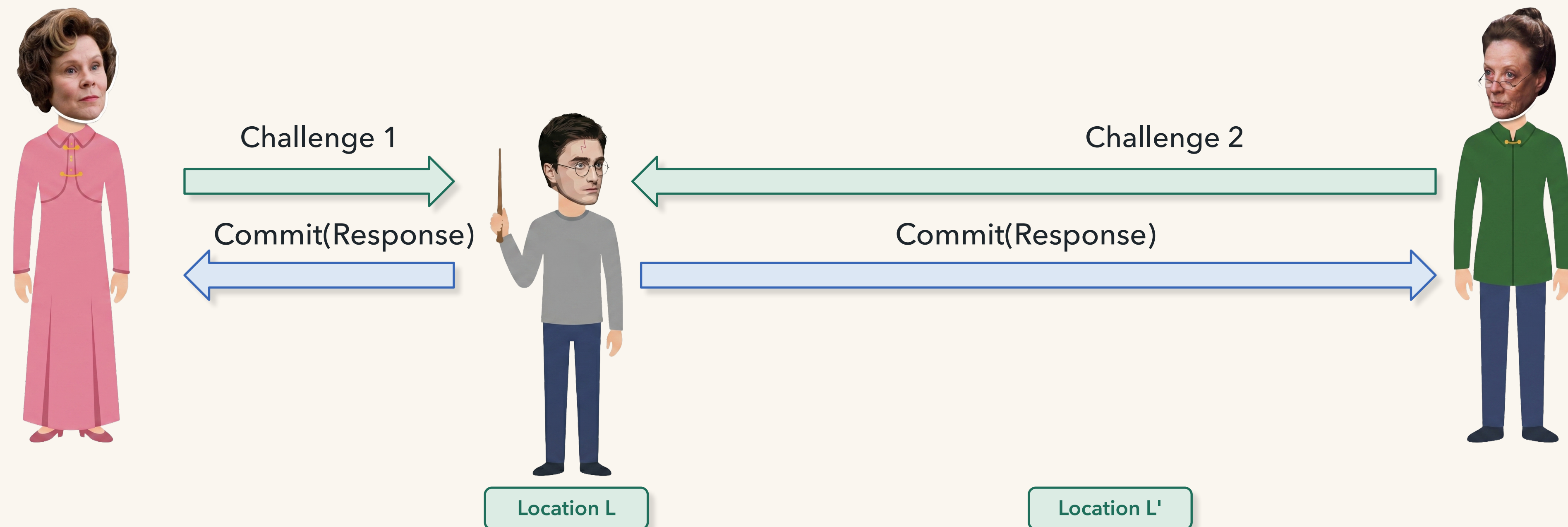
Black-Box Compiler: PV to Position Commitments

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Black-Box Compiler: PV to Position Commitments

Main idea: run *parallel instances* of PV for different locations, and *commit to responses*



Assume `commit()` is a classical, hiding commitment. Then the verifiers don't know if PV succeeded or failed!

Black-Box Compiler: PV to Position Commitments

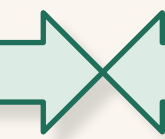
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Challenge 1



Location L



Challenge 2



Location L'

Black-Box Compiler: PV to Position Commitments

Main idea: run *parallel instances* of PV for different locations, and *commit to responses*



Challenge 1



Location L

Challenge 2

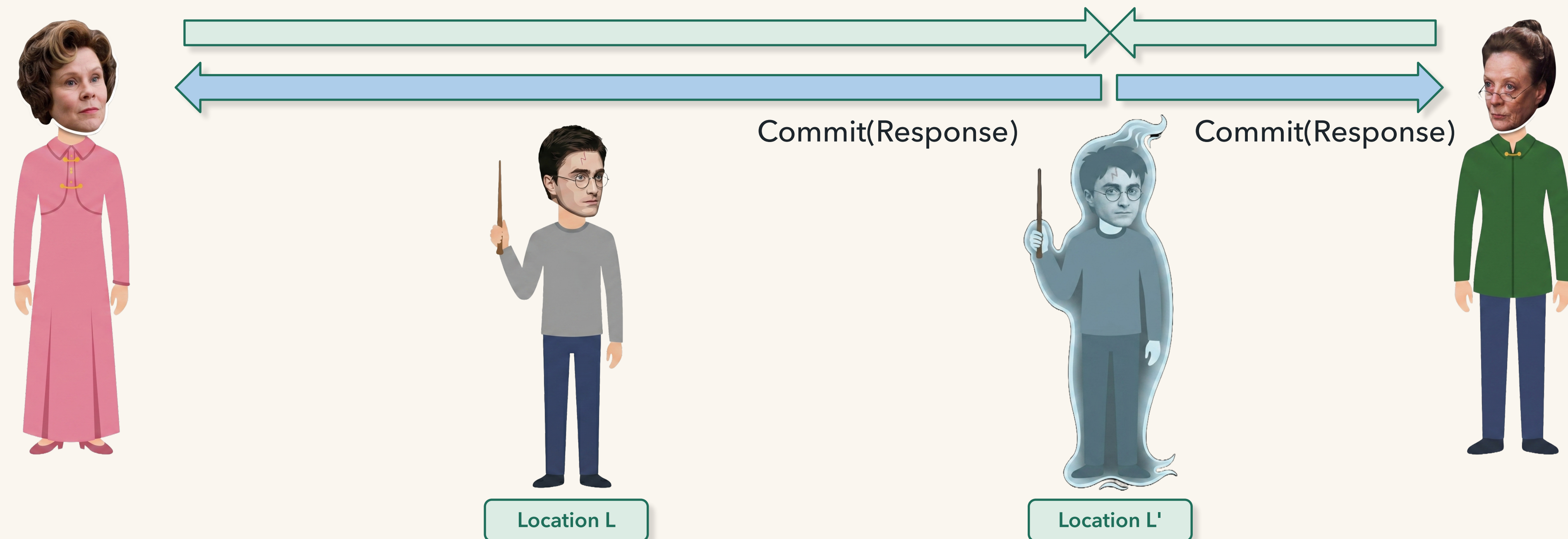


Location L'

Must pretend that there's a Peter at location L' running the same algorithm...

Black-Box Compiler: PV to Position Commitments

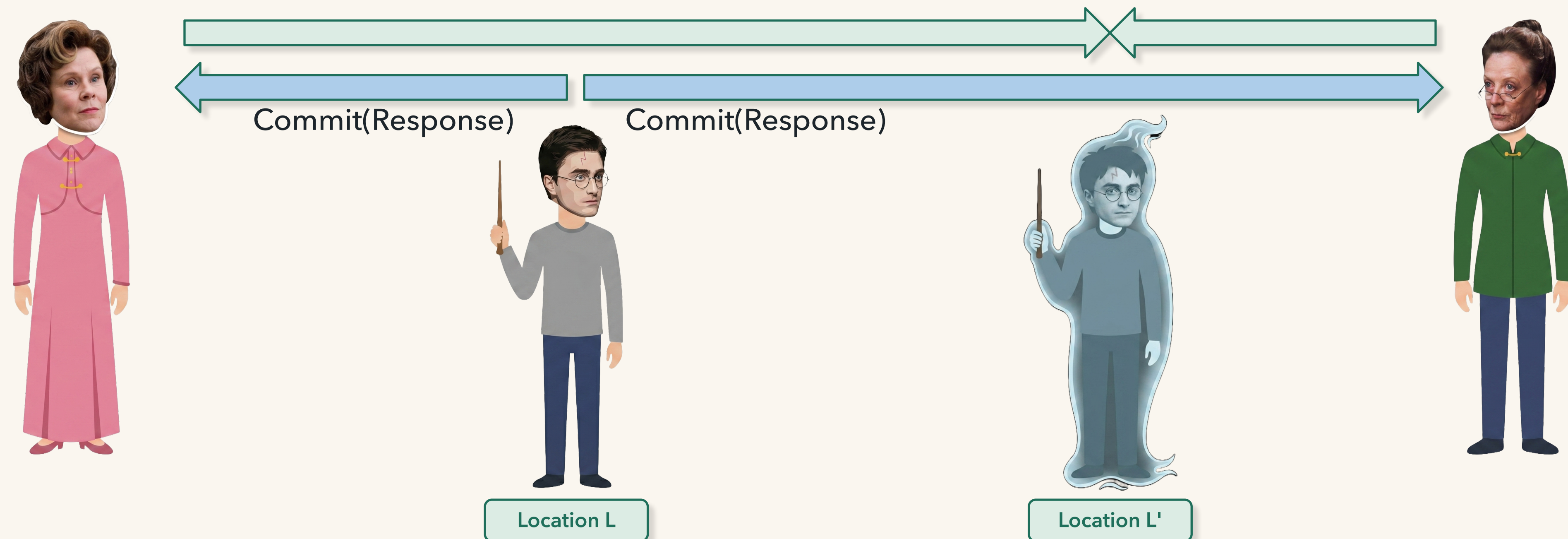
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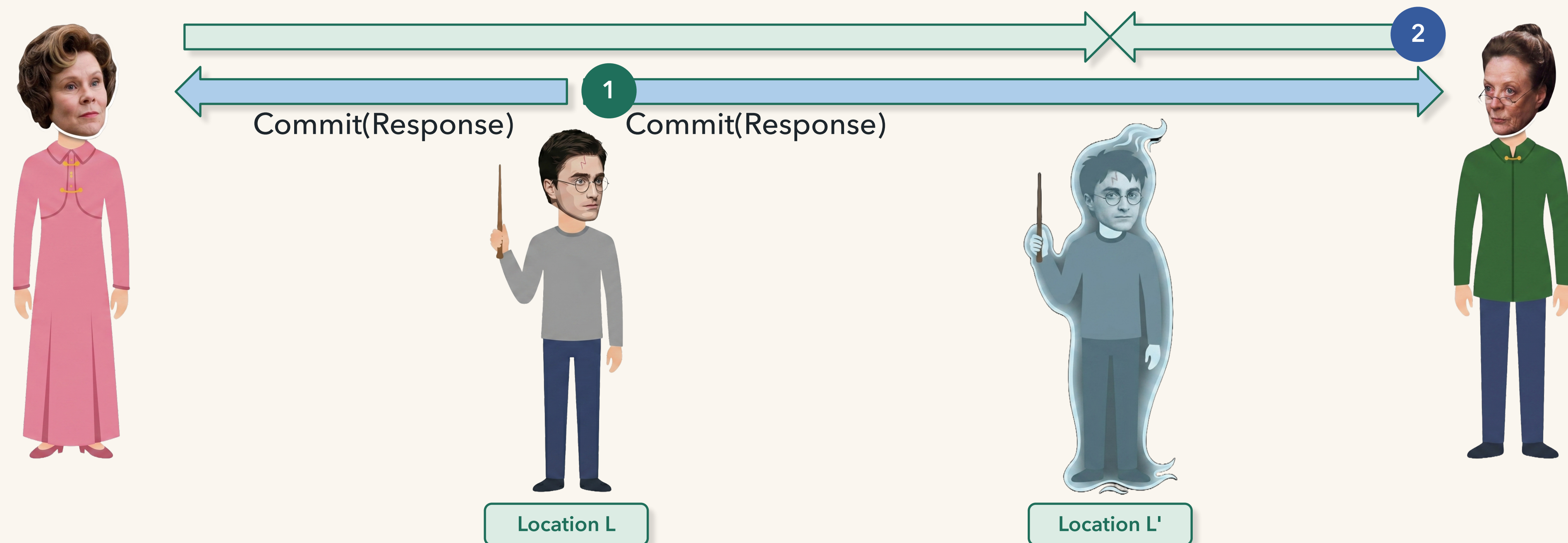
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Black-Box Compiler: PV to Position Commitments

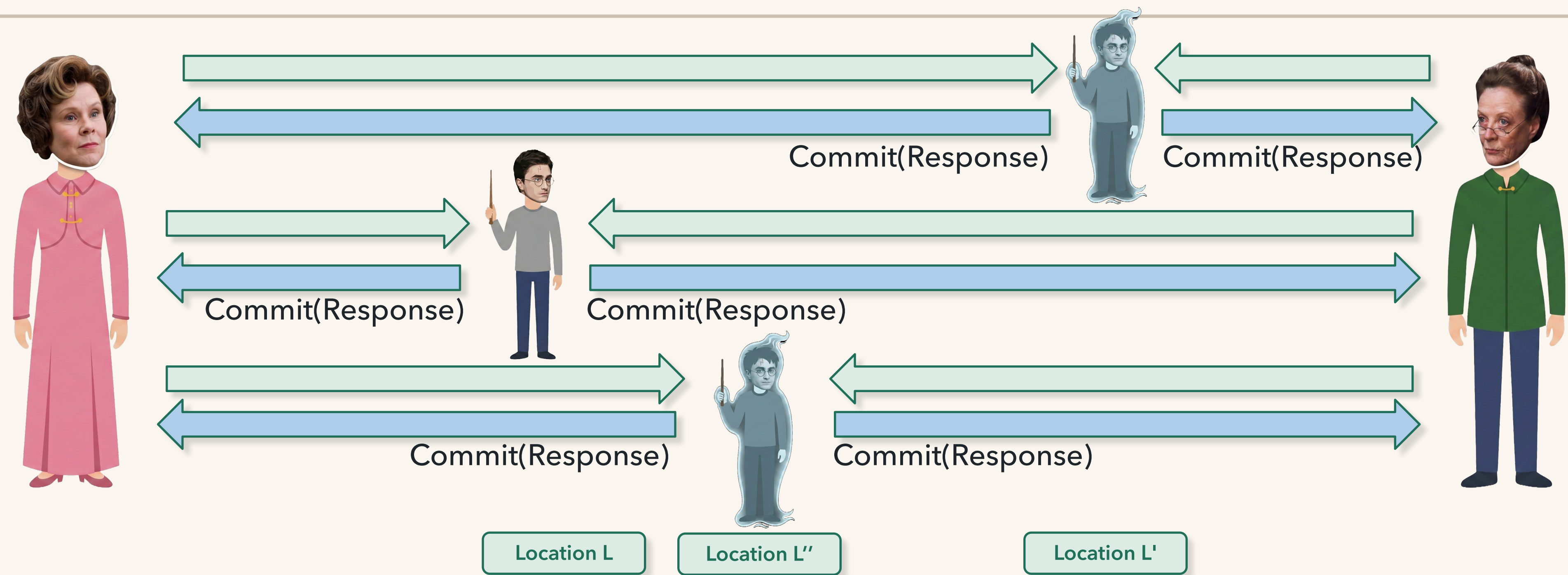
Main idea: run *parallel instances* of PV for different locations, and *commit to responses*



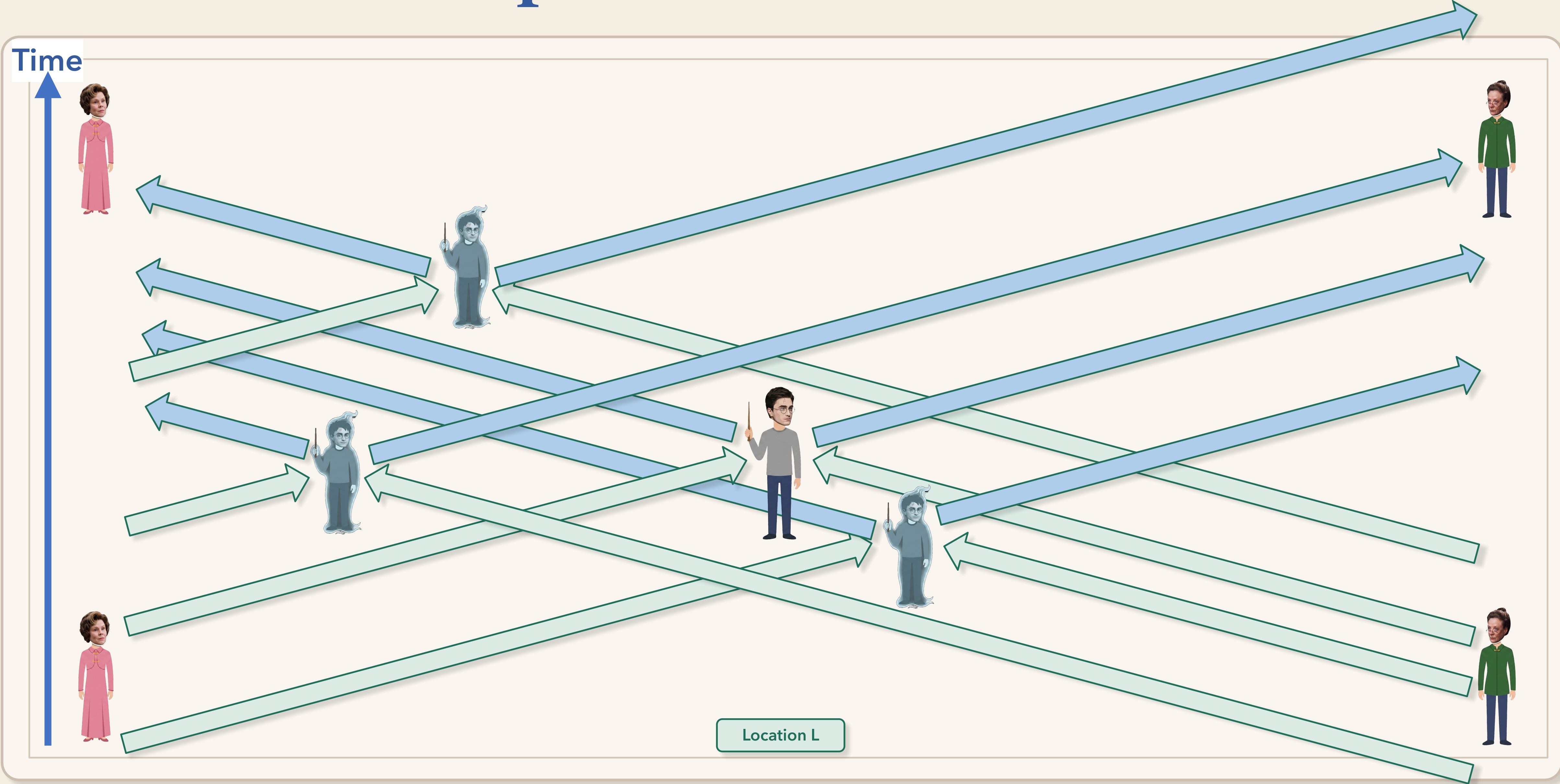
Notice (1) happens before (2). In other words, Peter needs a **head start** on the verifiers.

Black-Box Compiler: PV to Position Commitments

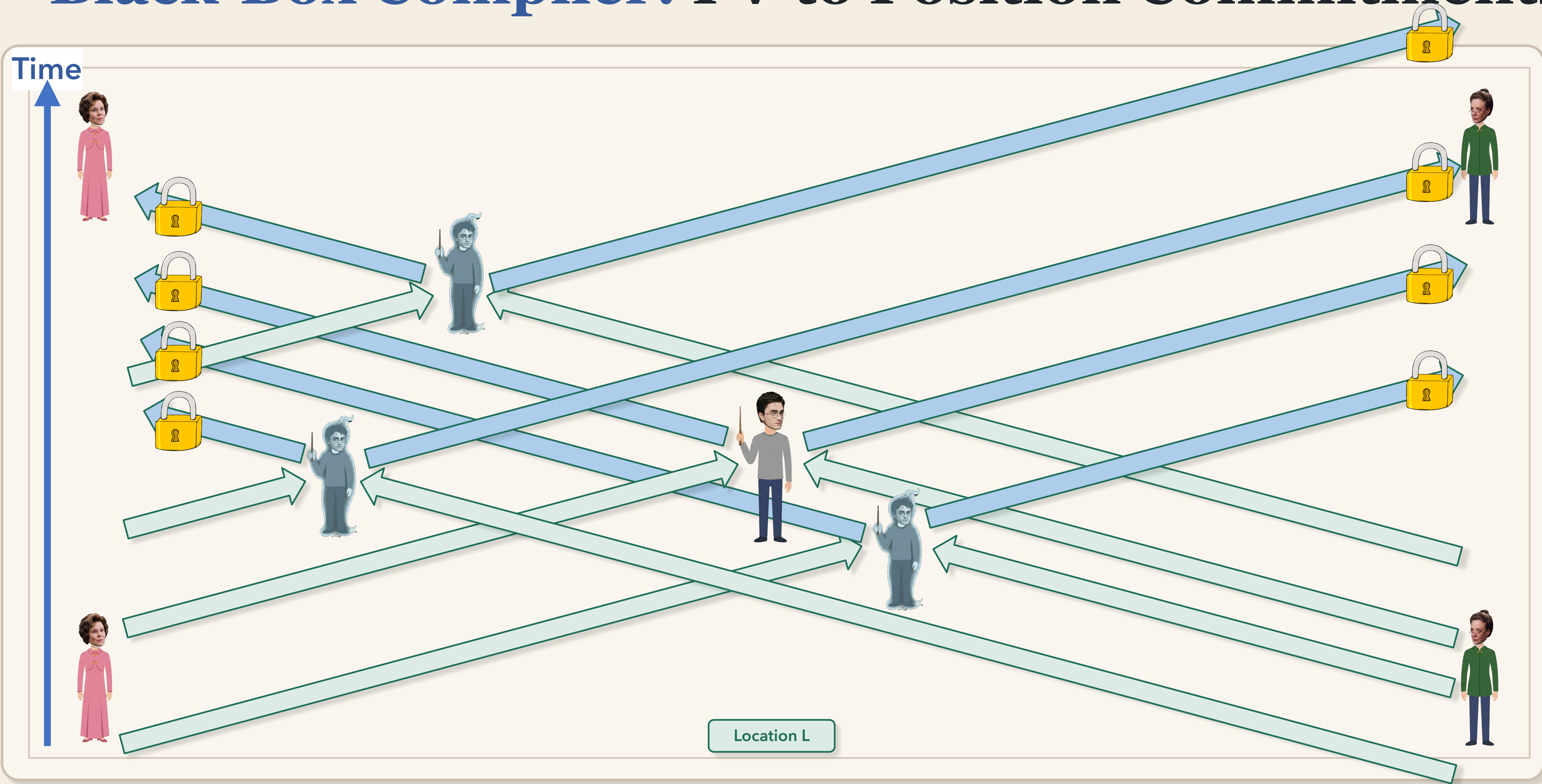
Main idea: run *parallel instances* of PV for different locations, and *commit to responses*



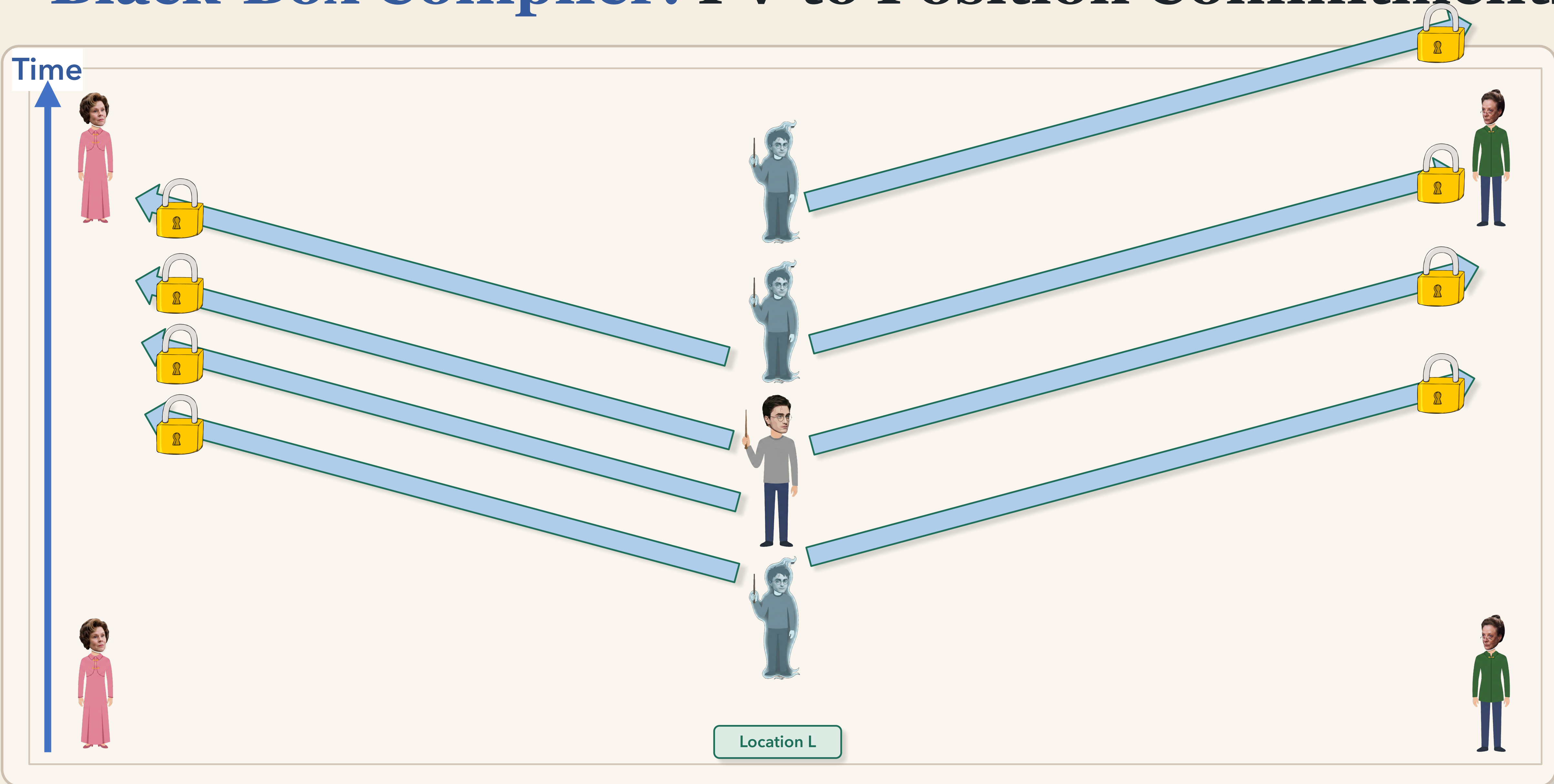
Black-Box Compiler: PV to Position Commitments



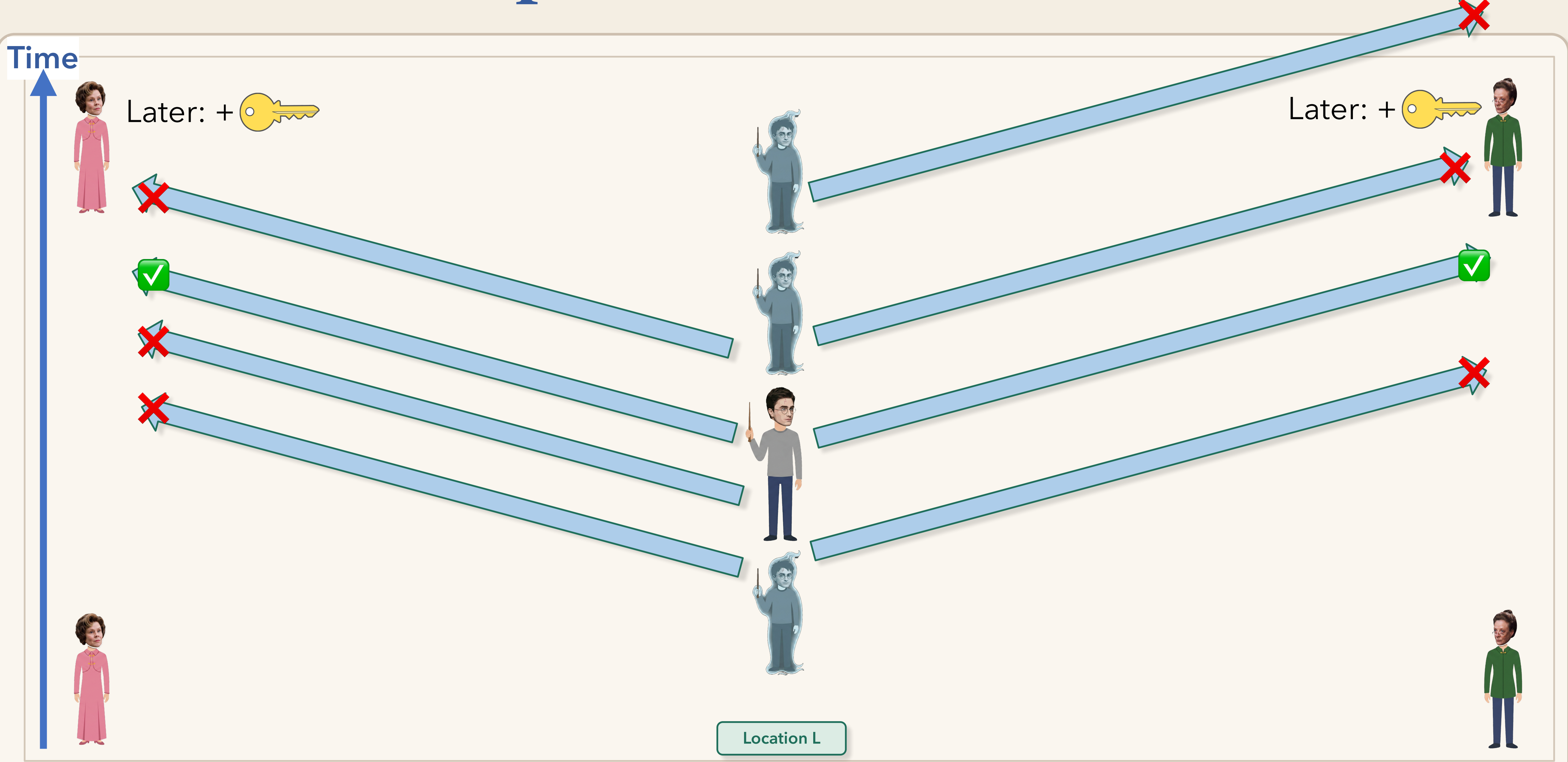
Black-Box Compiler: PV to Position Commitments



Black-Box Compiler: PV to Position Commitments



Black-Box Compiler: PV to Position Commitments



Security of our Compiler

Computational Hiding

- Verifiers always see a stream of commitments.
- One per potential location.
- Each message reveals nothing, by hiding property of the classical commitment scheme

Statistical Position Binding

- Assume the response commitments are statistically binding
- Then, prover must have had correct PV responses at the time of committing.
- From there, reduce to PV game.

Reduction is tricky, uses quantum advice!

From Position Commitments to ZK-PV

The final ZK-PV proof will proceed in two phases.

1 **Commitment**

Peter first *commits* to his location using a hiding and binding *position commitment* scheme. Call the resulting commitment C .

2 **Reveal a Property in Zero-Knowledge**

For any region R of choice, Peter then proves to Veronica, the following statement, using a standard zero-knowledge proof system for NP:

"There *exists* a reveal string r , such that $\text{Open}(C, r)$ gives some $(L, t) \in R$ "

Recap: Main Result

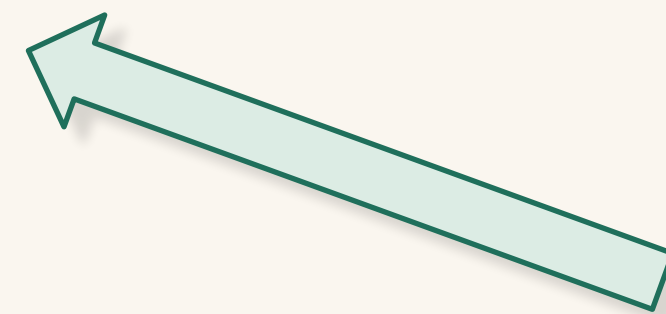
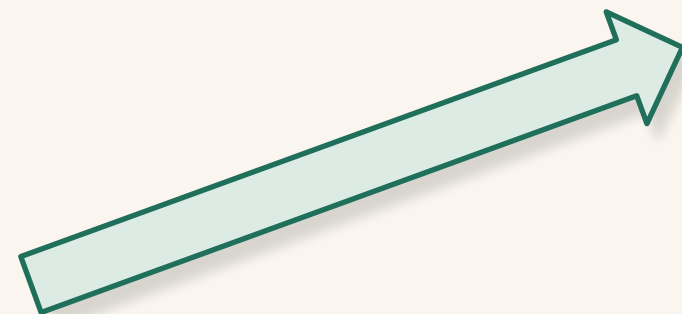
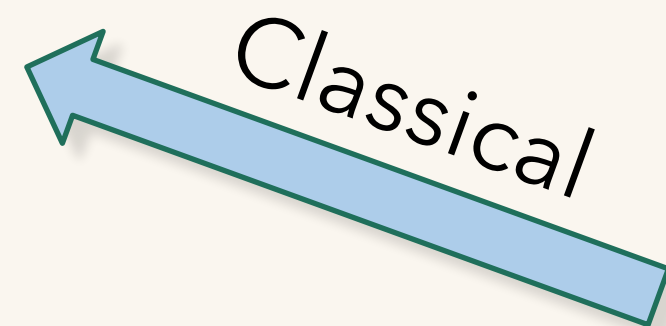
Assuming post-quantum secure one-way functions,
any* position verification protocol can be upgraded to a
zero-knowledge** position verification protocol,
through a *black-box* transformation.

*As long as it is single-round, with classical outputs

**Honest-verifier zero knowledge.
We'll talk more about this later.

Generic For All “Nice” PV Protocols

The PC compiler works on any classical-output, “single-shot” PV protocol.

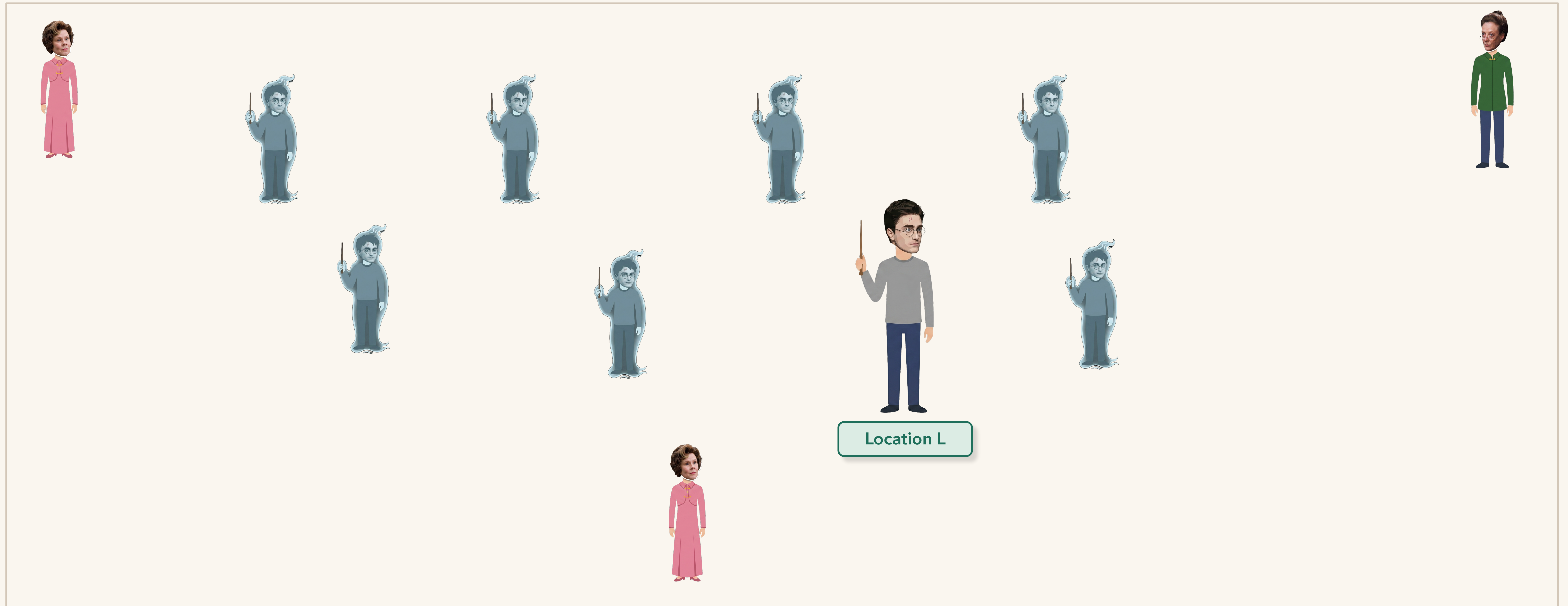


Examples of nice PV protocols:

- f-BB84
- Classical challenge-response
 - Secure in a no-colluding-provers model
- Classical-verifier protocol of [LLQ'22]
 - Secure assuming subexp. LWE, bounded entanglement
- Protocol of [CGMO'09]
 - Secure in the bounded-retrieval model
 - Bonus: prover can delay outputs

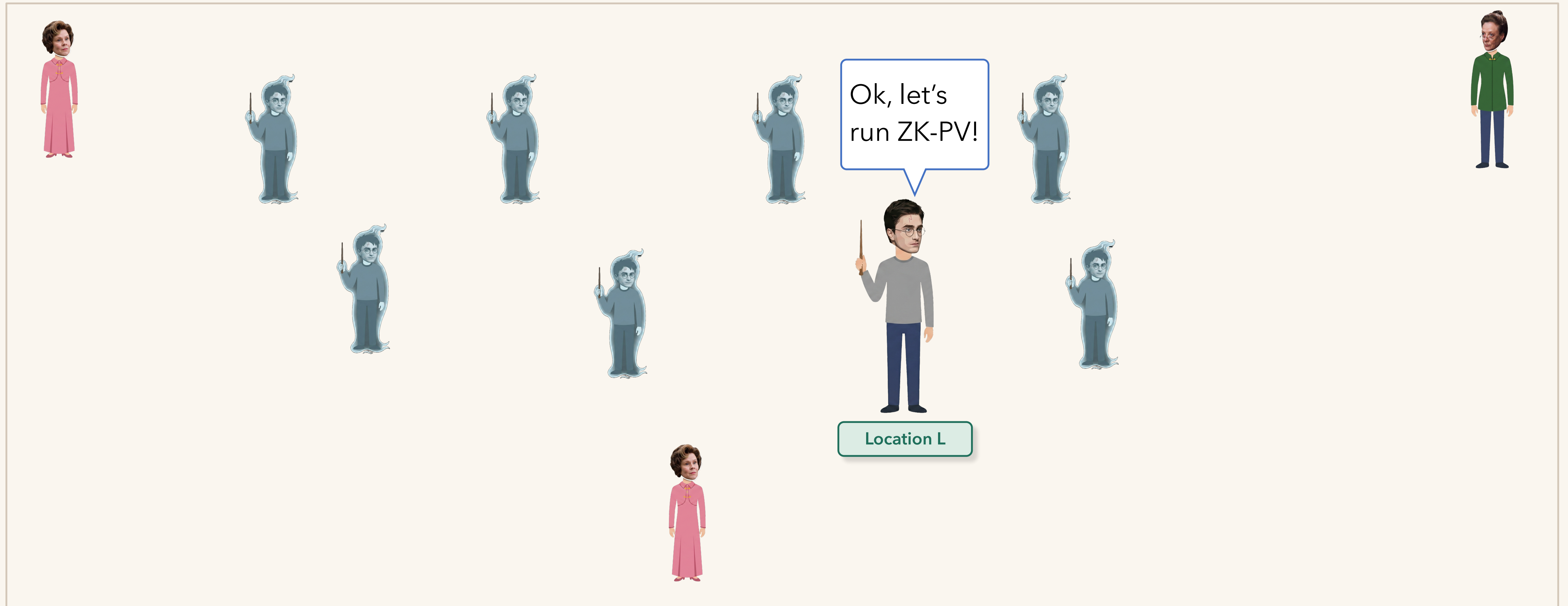
Honest Verifier Assumption

Usually, ZK against honest verifiers -> ZK against malicious verifiers. Why not here?



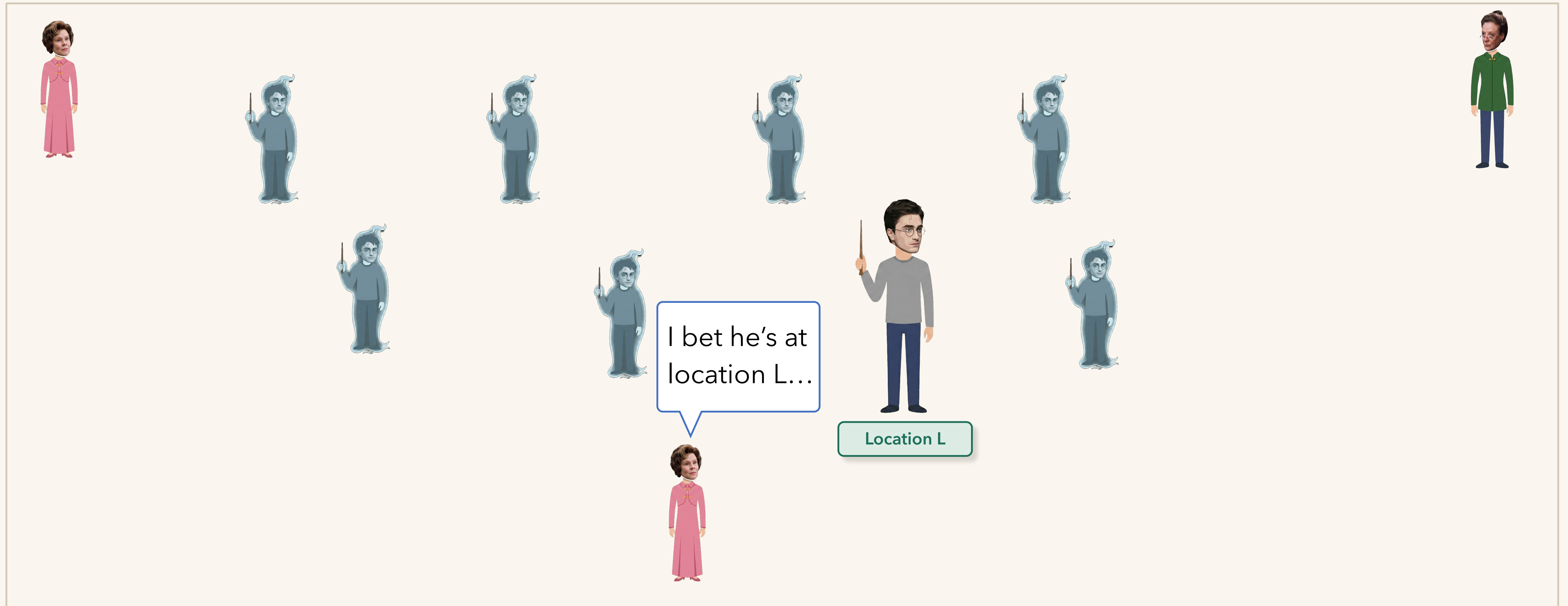
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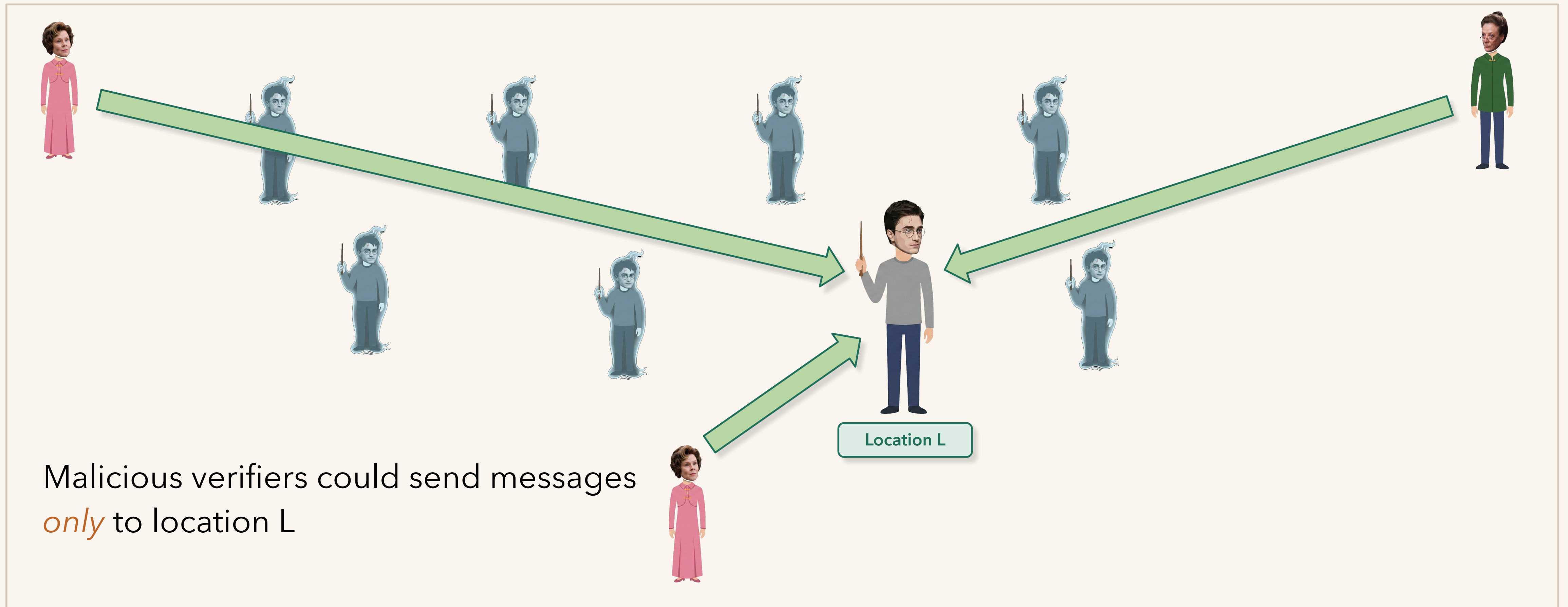
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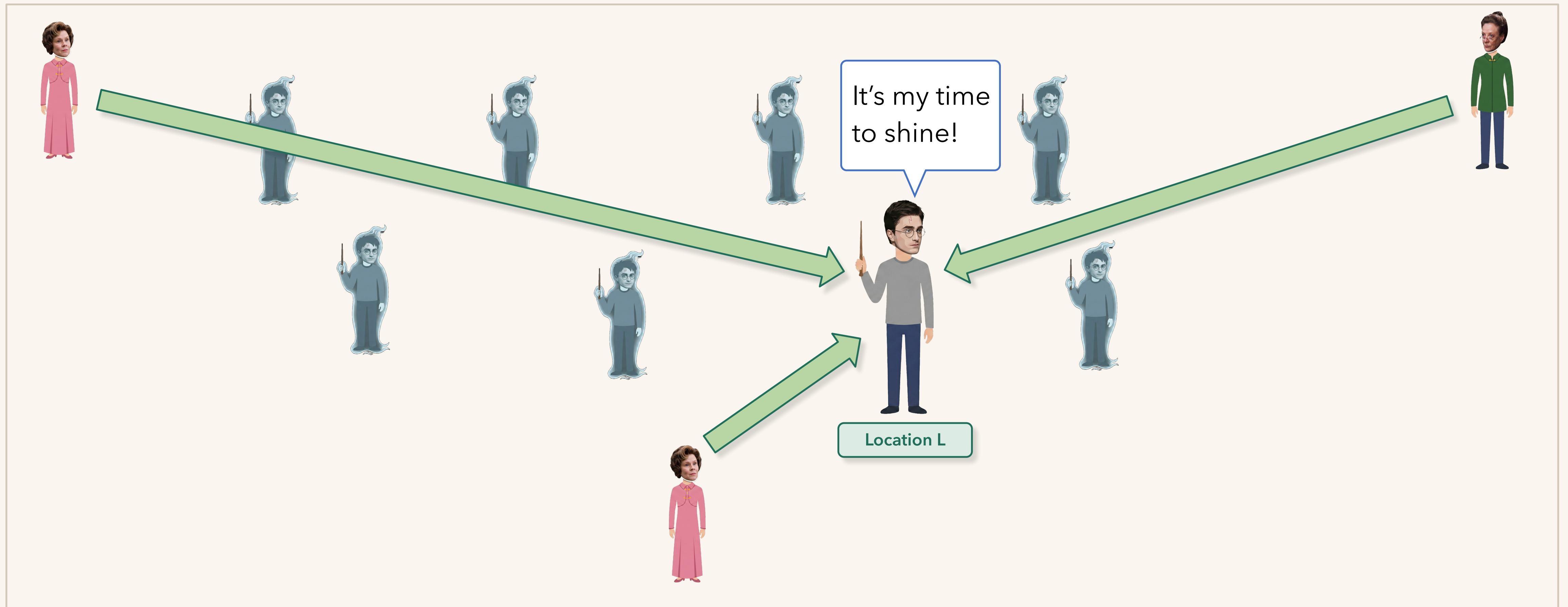
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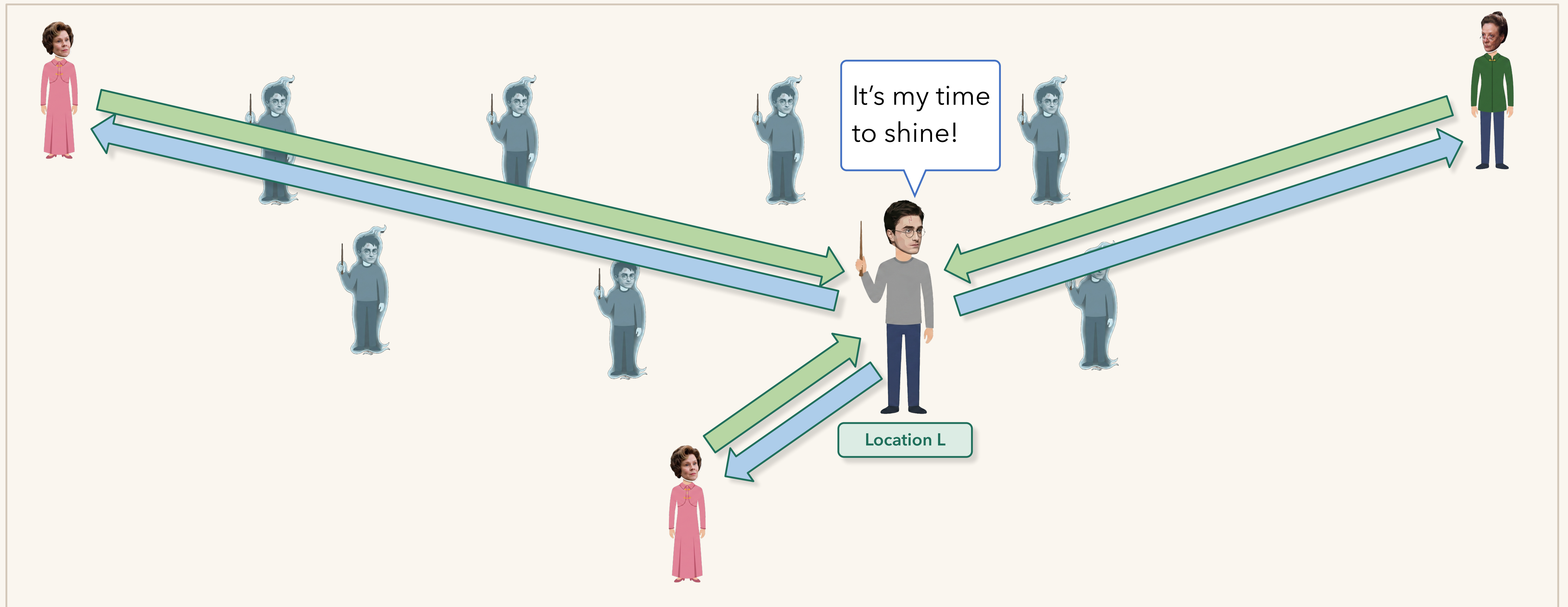
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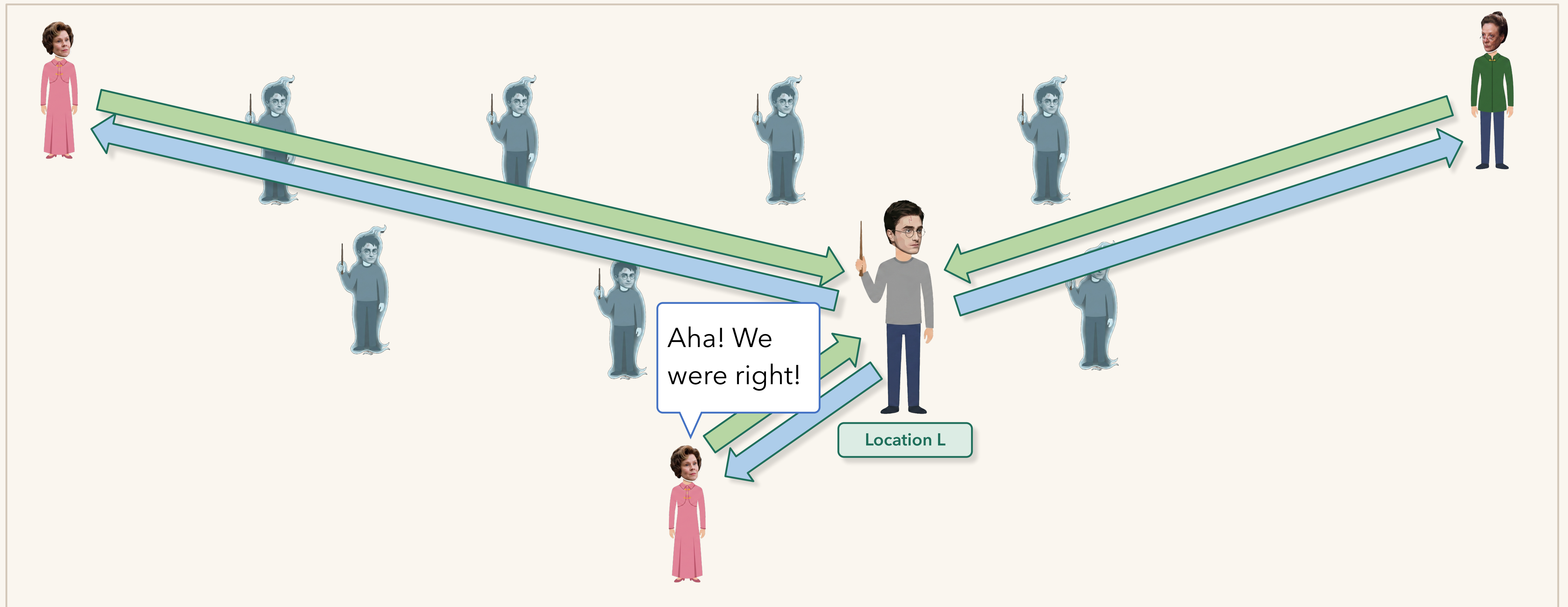
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Honest Verifier Assumption

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Final Recap

Our work:

1 Defining Zero-Knowledge Position Verification

- Previously, only *single-point* position verification protocols were considered.
- We extend this to **general spacetime regions**, and consider **privacy**.

2 Construct ZK-PV with a Black-Box Compiler From PV and OWF

- In other words, if we have secure position verification protocols and private key cryptography, then we can make those protocols private!

3 Not mentioned in this talk: Equivalence with PV+OWF, optimized constructions

Future Directions

- Can we relax idealistic modeling assumptions? E.g. perfect clocks/instantaneous computation
- Can we get security against malicious verifiers by adding stronger assumptions to the model?
- In this generalized setting, can we *link* together multiple proofs?
 - i.e. I was the same prover who proved “I was in the USA yesterday”, as the one who proved “today I am in Canada”.

Thank You!

Questions?